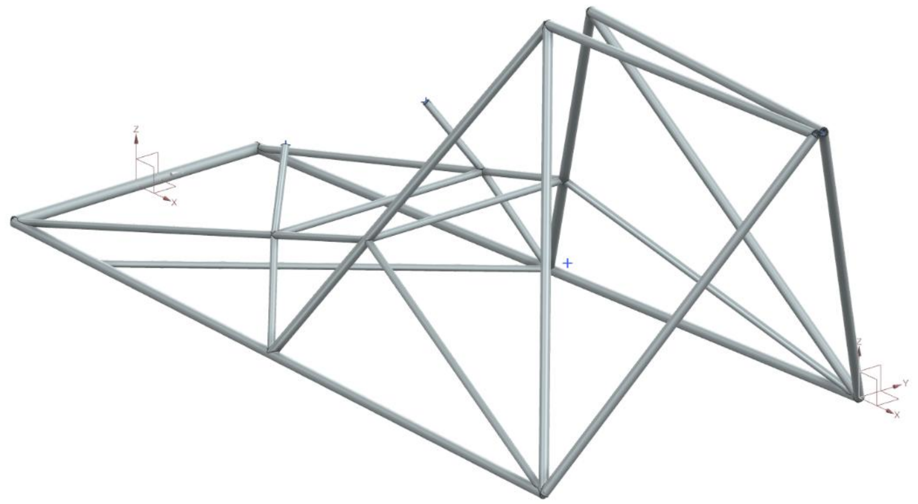
“AERODYNAMIC VEHICLE:

ULTRALIGHT HELICOPTER (SINGLE- SEATER) ”

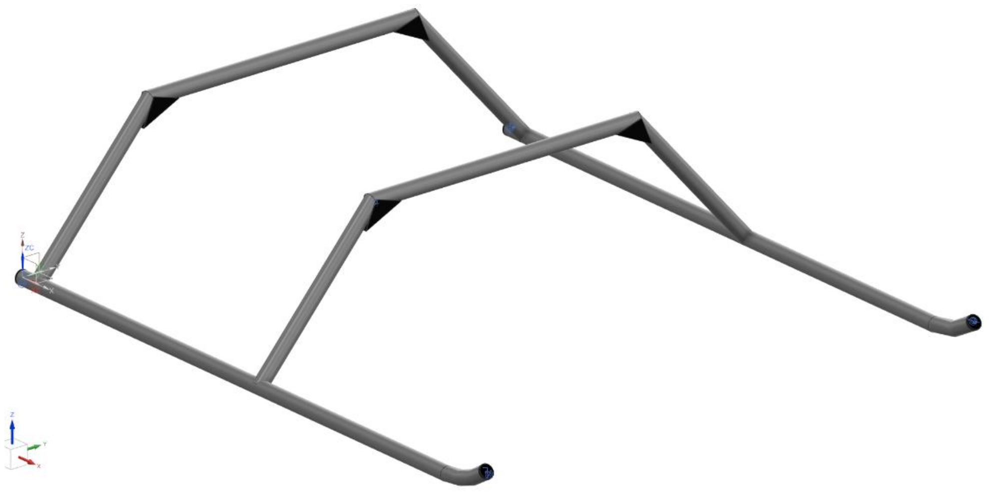
CAD Geometry Files Catalogue

1. Primary Structure



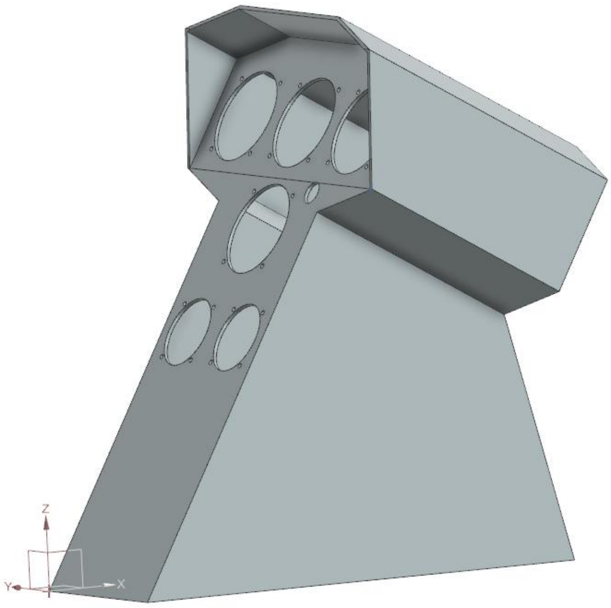
Material: 4130 Steel Tubing (Chromoly)
Triangulated Truss Design ensures primary loads are pure tension or compression rather than bending moments. 4130 is used due to its excellent strength-to-weight ratio and offers great fatigue resistance maintaining integrity under TIG welding. Steel has great inherent damping characteristics absorbing harmonic vibrations from rotors.

2. Landing Gear Skids



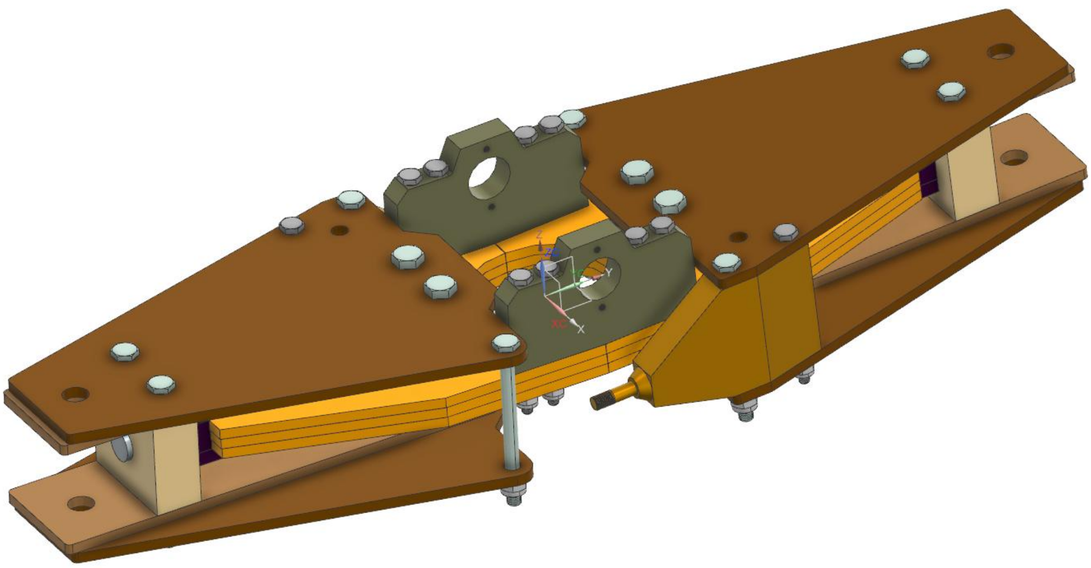
Material: 4130 Steel
The base structure is supported by 4 Reinforcements at the joints and the open ends are closed by 4 Skids Caps which are made of Aluminum reducing air drag.

3. Instrumental Panel

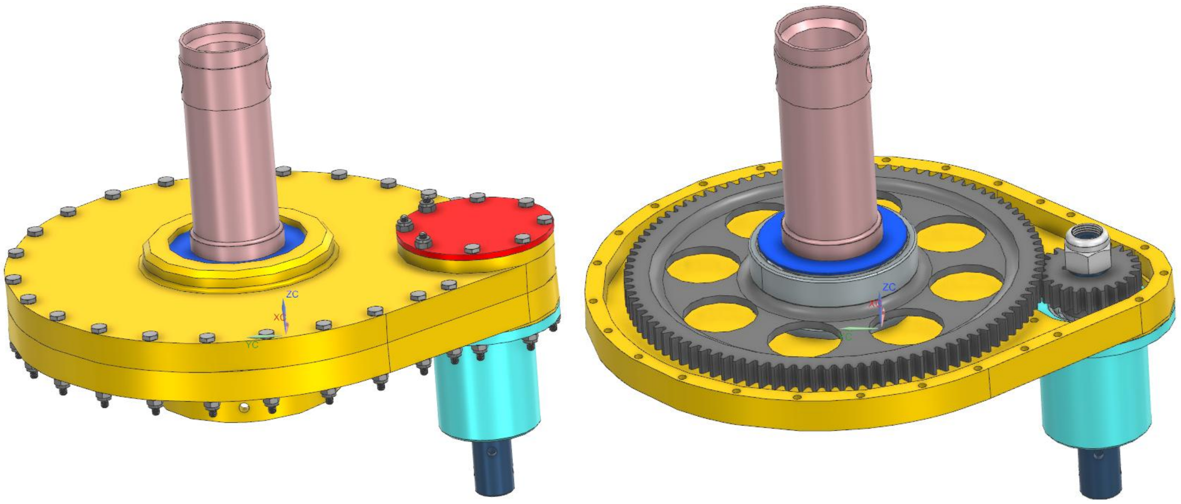


Material: 6061 T6 Aluminum
The cutoffs are pre-machined mounting points for Altimeter, R.P.M, Airspeed, Ignition Key, Turn and Bank, Cylindrical Head Temperature and Exhaust Gas Temperature (as proposed in document). The Ergonomic Incline helps in easy scanning during flights.

4. Main Rotor Head



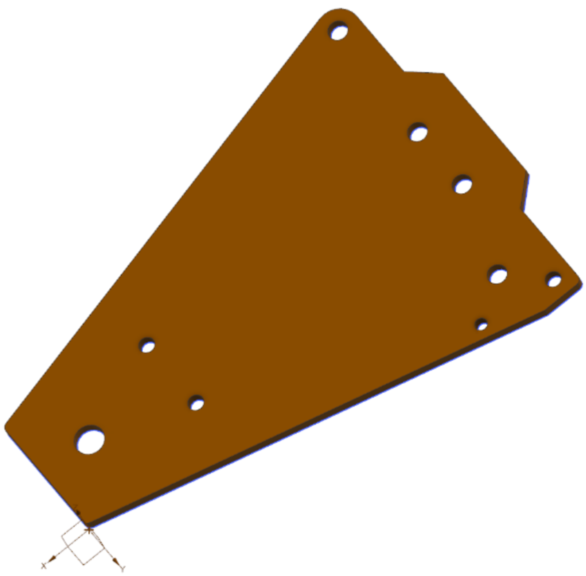
5. Main Rotor Gearbox Assembly



The upcoming section delineates the different subparts that were assembled together to build the various components and the final geometry ---

6. Main Rotor

i. Blade Grip



ii. Blade Grip Counterweight

Material: 1045 Steel

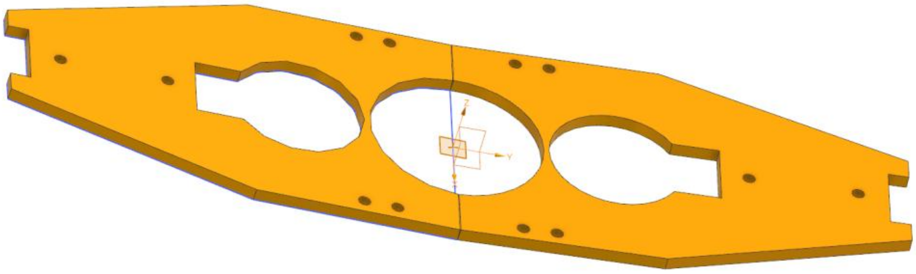
iii. Bushings

Material: Steel tube

Material: Steel

They reduce friction and protect against wear and tear of parts.

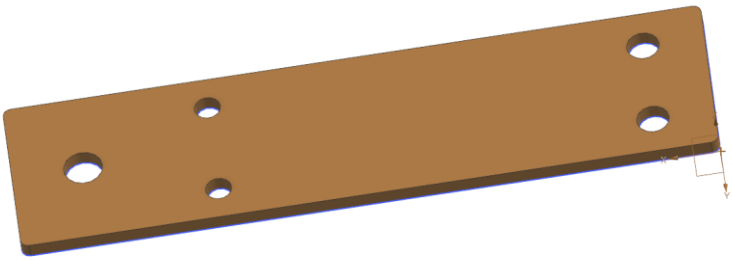
iv. Center Plate



Material: 2024- T3 Aluminum

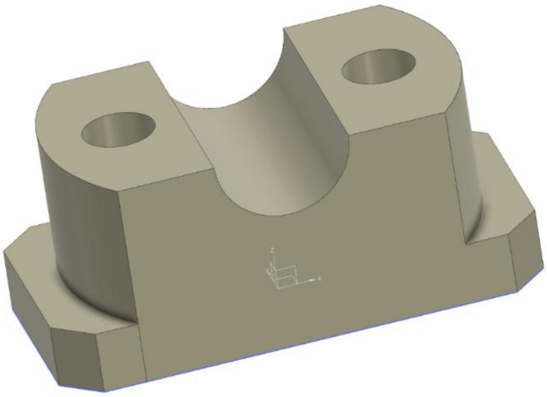
It is bent along the middle bending line by total 4 degrees.

v. Reinforcement Strap



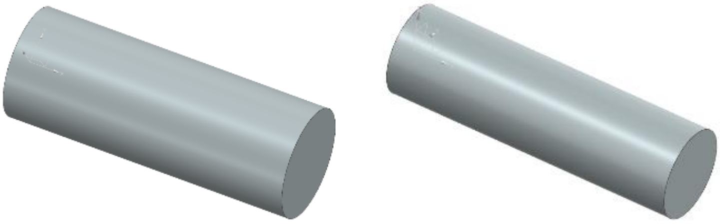
Material: 2024- T3 Aluminum

vi. Axle Grip



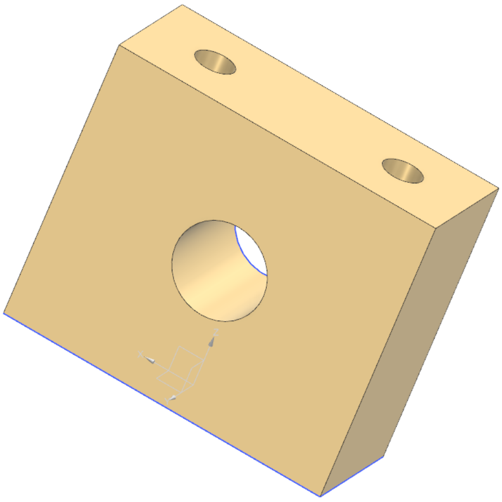
Material: 2024- T3 Aluminum

vii. Axles



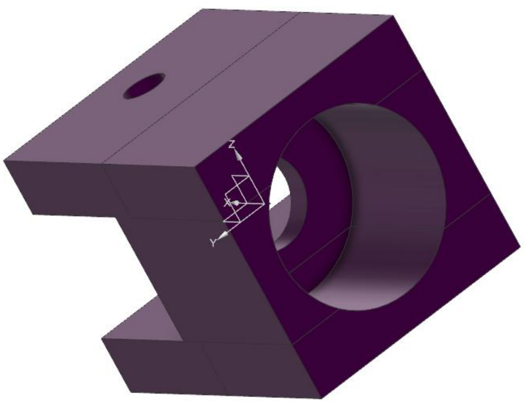
Material: 8620 Steel (Case Hardened)

viii. Axle Housing



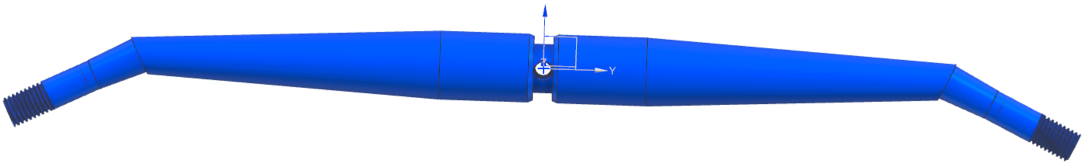
Material: 2024- T3 Aluminum

ix. Claw



Material: 4140 Steel with higher carbon content and UTS
It holds the plates together. Inside it is an H Bearing Housing.

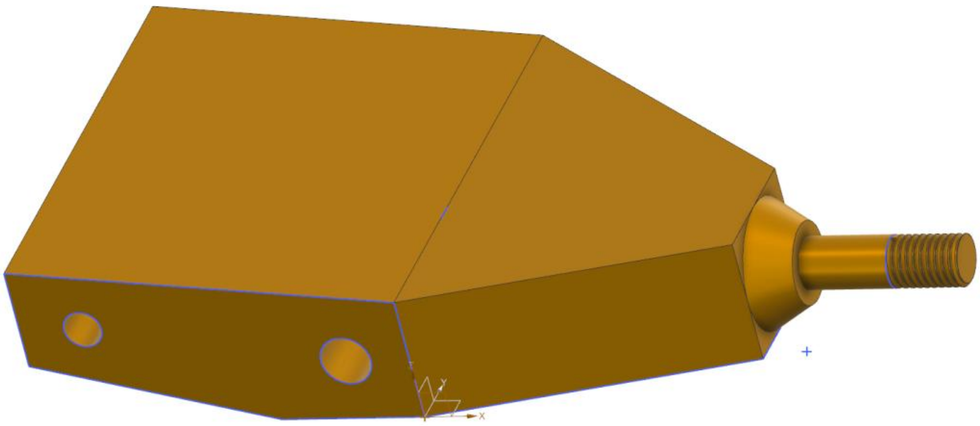
x. Blade Control Master Horn



Material: 1045 Steel
Internal threading at the centre hole (** 8-36 UNF)-
Pitch= 0.0278in/ 0.706mm
Major Diameter= 0.1640in/ 4.166mm
Minor Diameter=0.1339in/ 3.4012mm
Tap Drill= 0.1375in/ 3.50mm

External threading at two ends-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Shaft Diameter= 0.2464in/ 6.2593mm

xi. Blade Control Horn



Material: 2024- T3 Aluminum
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Shaft Diameter= 0.2464in/ 6.2593mm

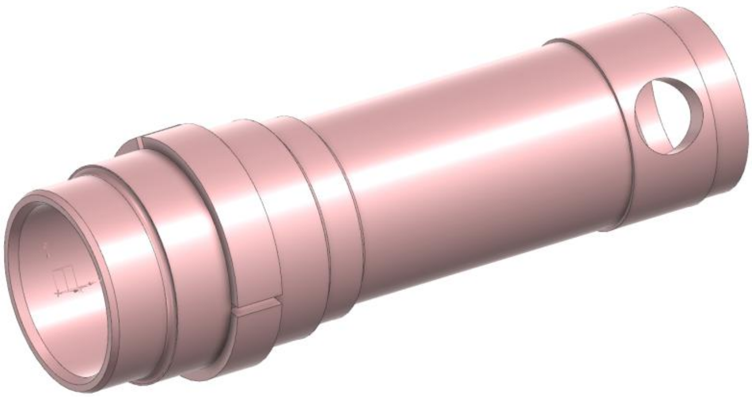
xii. Control Sliding Axle

Material: 4140 Steel

xiii. Ball

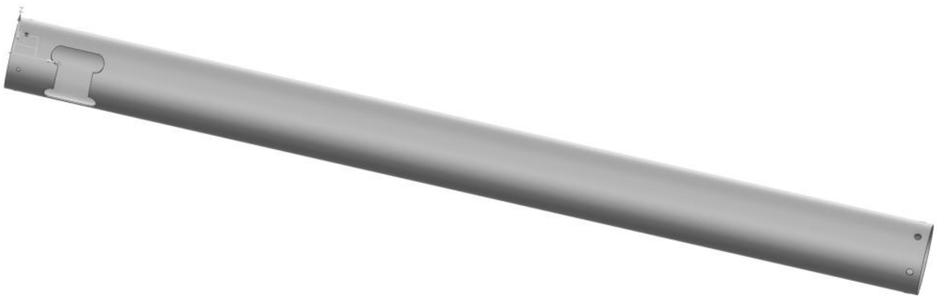
Materisl: 8620 Steel (Cases hardened)

xiv. Main Rotor Shaft



Material: 4140 Steel

xv. Main Rotor Mast



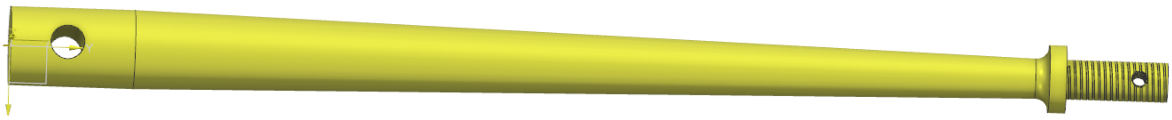
Material: 6061- T6 Aluminum Tube

xvi. Gearbox Attachment Ring

Material: 4130 Steel
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill- 0.213in/ 5.4mm

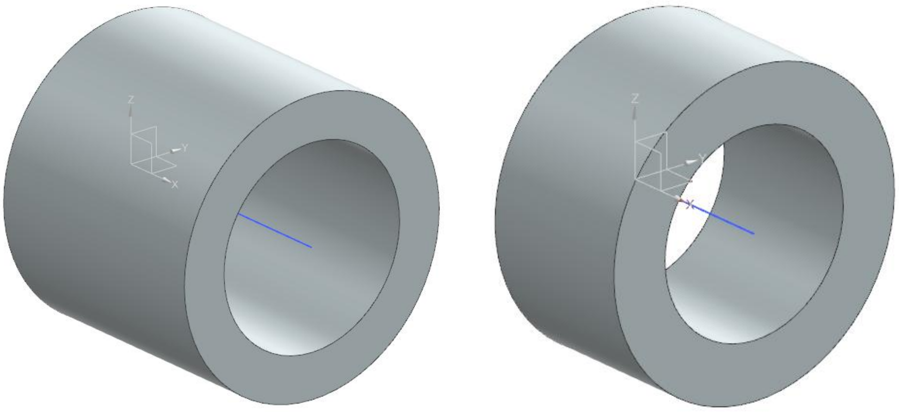
7. Control Systems

i. Push- Pull Axle



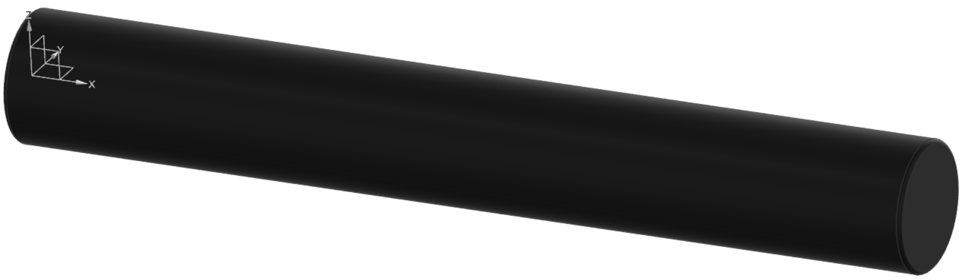
Material: 4140 Steel
Threading-
Pitch= 0.0417in/ 1.0583mm
Major Diameter= 0.3125in/ 7.9375mm
Minor Diameter= 0.2674in/ 6.7918mm

ii. Bushings



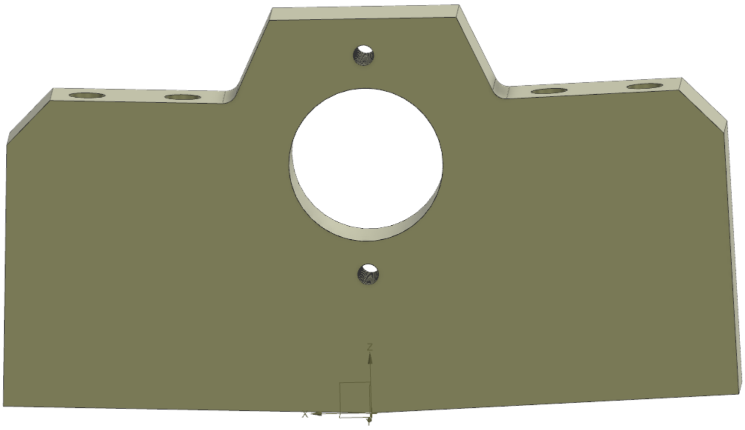
Material: Steel

iii. Main Rotor Axle



Material: 8620 Steel (Case Hardened)

iv. Pivot Head

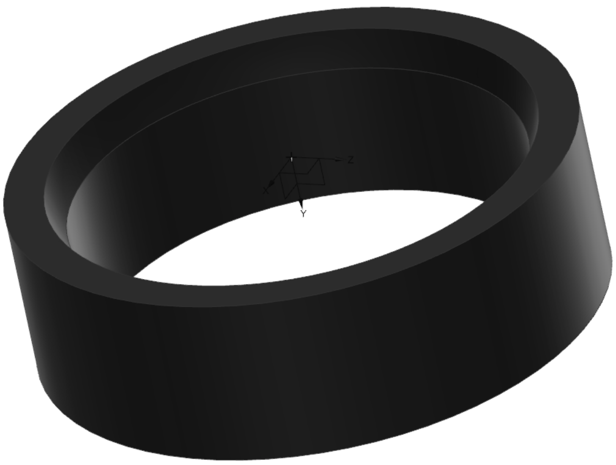


Material: 2024- T3 Aluminum
Threading-
Pitch= 0.03125in/ 0.79375mm
Major Diameter= 0.1140in/ 2.8956mm
Minor Diameter= 0.1040in/ 2.6416mm
Tap Drill= 0.1110in/ 2.8194mm

v. Race/ Nut

Material: 1050 Steel
Threading-
Pitch= 0.0833in/ 2.09mm
Major Diameter= 1.3750in/ 34.925mm
Minor Diameter= 1.2848in/ 32.6346mm
Shank Diameter= 1.3667in/ 34.7133mm

vi. Ball Race

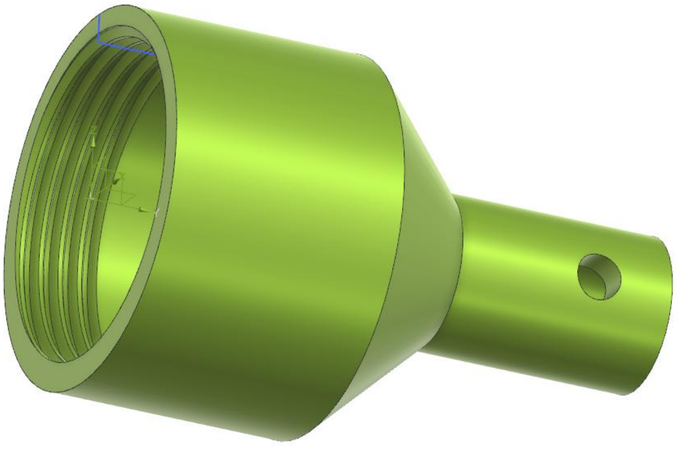


Material: 1050 Steel

vii. Ball/ Bearing Housing

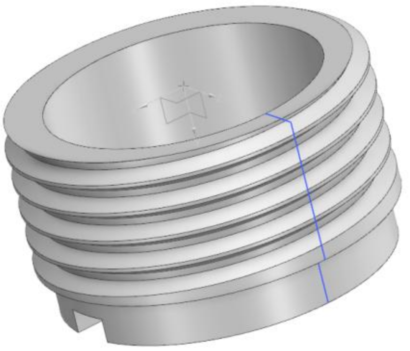
Material: 1050 Steel
Pitch= 0.0714in/ 1.8143mm
Major Diameter= 0.875in/ 22.225mm
Minor Diameter= 0.7977in/ 20.2616mm
Tap Drill= 0.8666in/ 22.00mm

viii. Central Uniball Housing



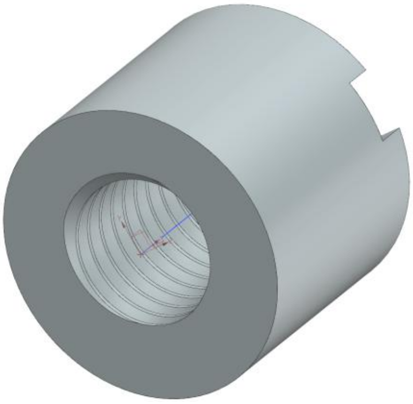
Material: 1045 Steel
Threading-
Pitch= 0.0833in/ 2.09mm
Major Diameter= 1.3750in/ 34.925mm
Minor Diameter= 1.2848in/ 32.6346mm
Tap Drill= 1.299in/ 33mm

ix. Nuts



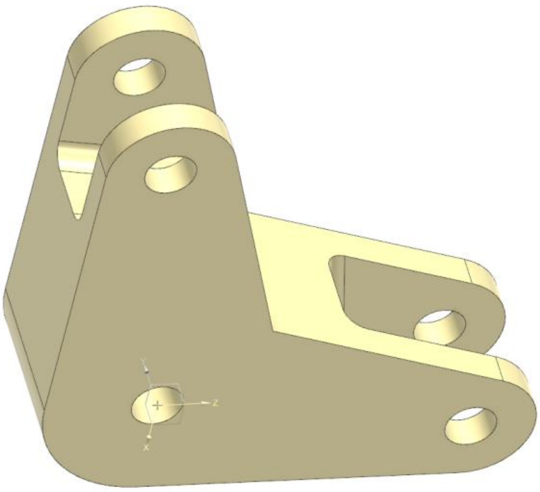
A.
Material: 1045 Steel

Pitch= 0.0714in/ 1.8143mm
Major Diameter= 0.875in/ 22.225mm
Minor Diameter= 0.7977in/ 20.2616mm
Shank Diameter= 0.875in/ 22.225mm



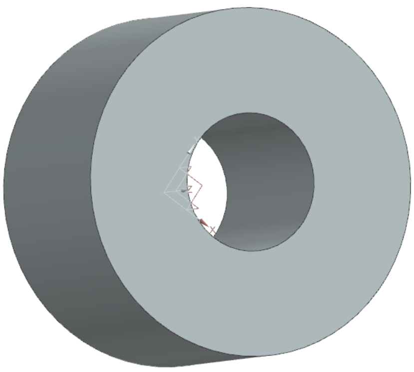
B.
Material: 1045 Steel
Pitch= 0.04167in/ 1.0583mm
Major Diameter= 0.3125in/ 7.9375mm
Minor Diameter= 0.2674in/ 6.7918mm
Tap Drill= 0.2720in/ 6.9mm

x. Cyclic Upper Bellcrank



Material: 2024- T3 Aluminum

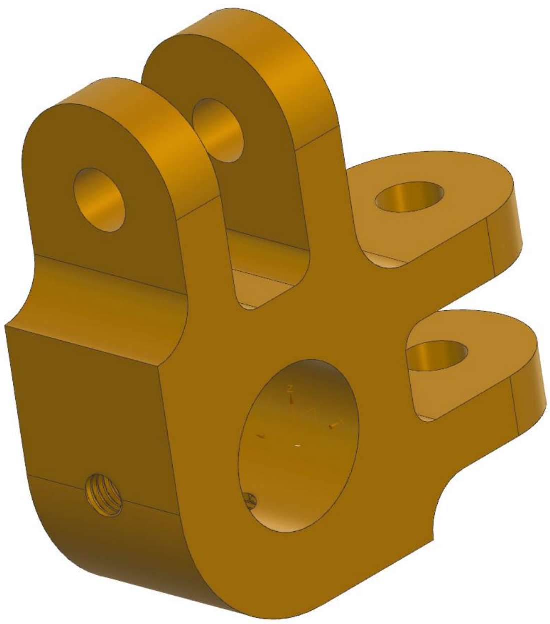
xi. Bushings- Spacer



Material: 2024- T3 Aluminum

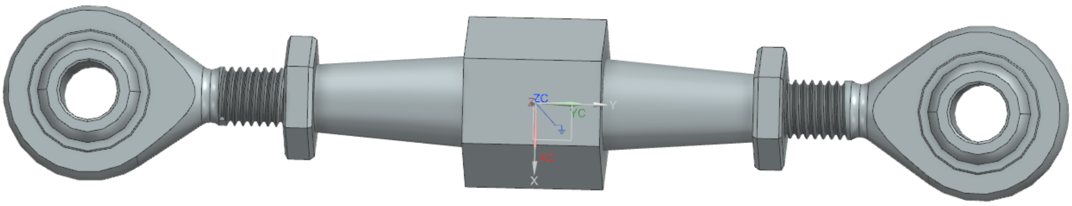
Material: Steel/ Brass

xii. Cyclic/ Collective Central Link



Material: 2024- T3 Aluminum
Threading-
Pitch= 0.03125in/ 0.79375mm
Major Diameter= 0.19in/ 4.826mm
Minor Diameter= 0.1562in/ 3.9667mm
Tap Drill= 0.159in/ 4.10mm

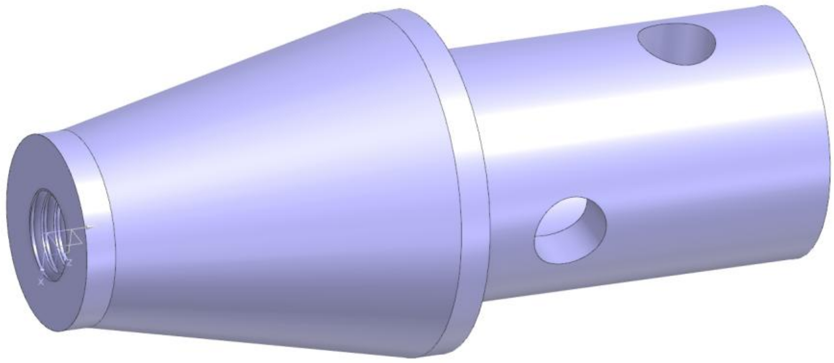
xiii. Blade Control Linkage



Material (Central Body): 2024- T3 Aluminum
Material (HMVV-4 and HMLVV-4 Rod Ends): Carbon Steel
Material (AN316-4R and AN316-4L): Cadmium plated Carbon Steel
Opposite (right hand and left hand) threading is used at both end such that on twisting the centre body by spanner ensures simultaneous outward or inward movement of the rod ends. The hex-nuts serve for jamming.
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm

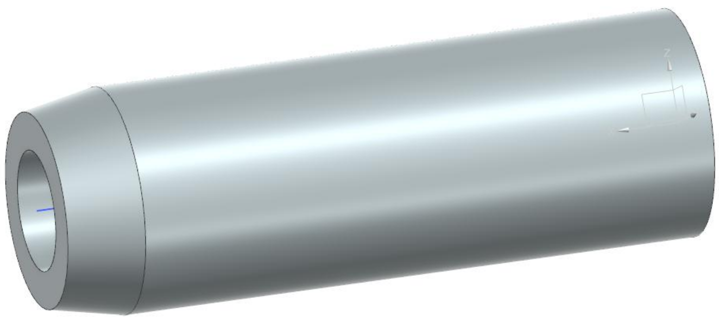
Minor Diameter=0.2113in/ 5.367mm
Shank Diameter= 0.25in/ 6.35mm
Tap Drill= 0.213in/ 5.4102mm

xiv. Push Pull Tubes Threaded Ends



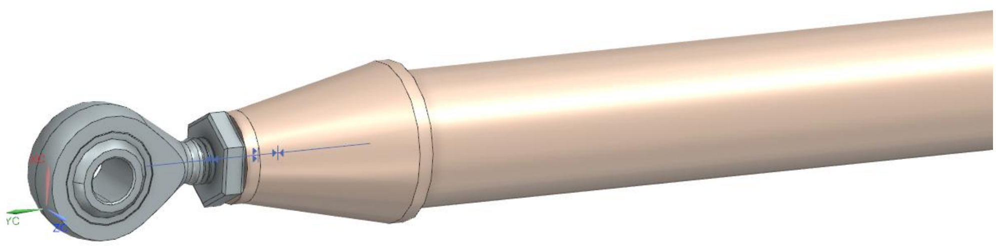
Material: 2024- T3/ 6061- T6 Aluminum
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill= 0.213in/ 5.4102mm (Right and left both)

xv. Drag Brace Bushing



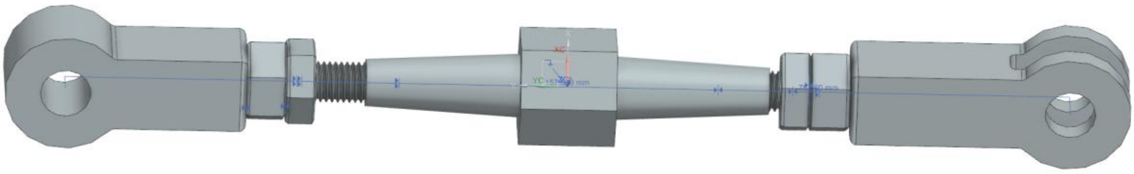
Material: 2024- t3 Aluminum

xvi. Central Push Pull Tube



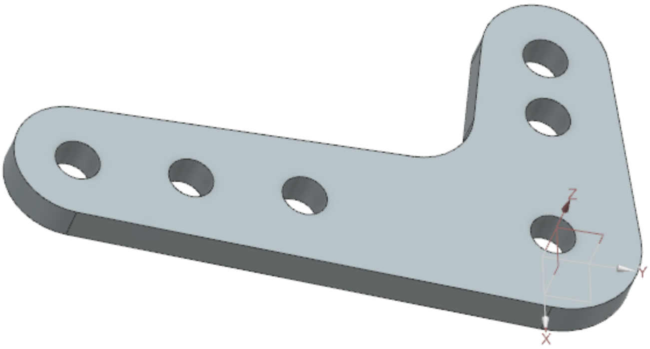
Material: 4130 Steel Tube
HVV-4 Rod End and AN316-4R jamming nut have been used (refer xiii for threading).

xvii. Blade Drag Brace



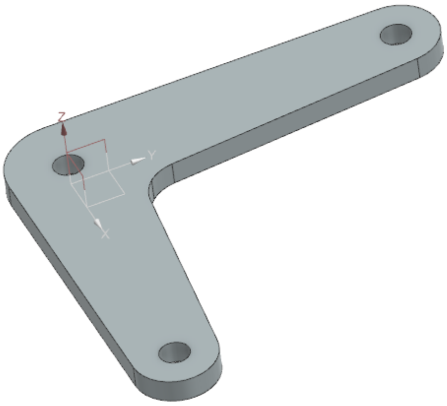
Material: Steel (refer xiii for threading)

xviii. Cyclic Control Lower Bellcrank



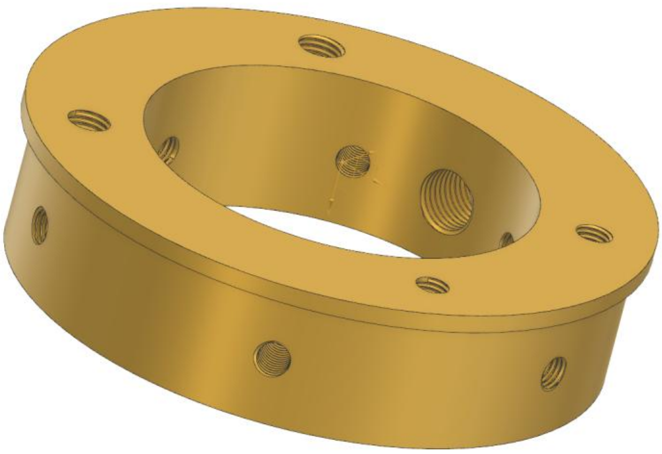
Material: 2024- T3 Aluminum

xix. Collective Control Lower Bellcrank



Material: 2024- T3 Aluminum

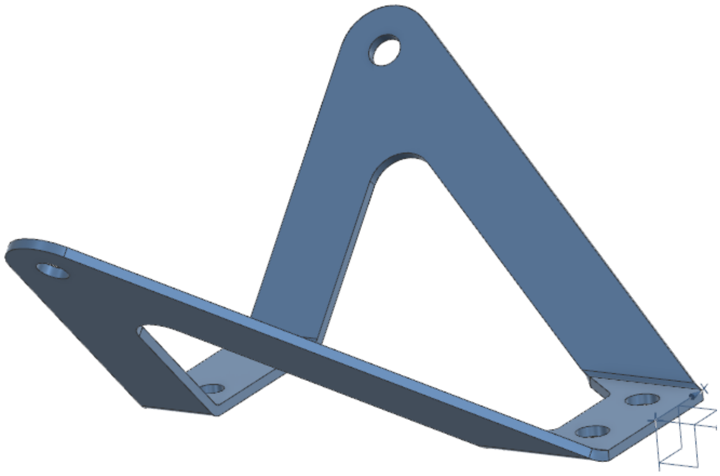
xx. Bellcranks Mounting Ring



Material: Aluminum
Threading-

a) Pitch= 0.0357in/ 0.907mm
   Major Diameter= 0.2500in/ 6.350mm
   Minor Diameter=0.2113in/ 5.367mm
   Tap Drill- 0.213in/ 5.4mm
b) Pitch= 0.05in/ 1.27mm
   Major Diameter= 0.4375in/ 11.1125mm
   Minor Diameter=0.3834in/ 9.7384mm
   Tap Drill- 0.390in/ 9.91mm
c) Pitch= 0.0417in/ 1.0583mm
   Major Diameter= 0.3125in/ 7.9375mm
   Minor Diameter=0.2674in/ 6.7918mm
   Tap Drill- 0.2720in/ 6.9mm

xxi. Central Bracket



Material: 4130 Steel

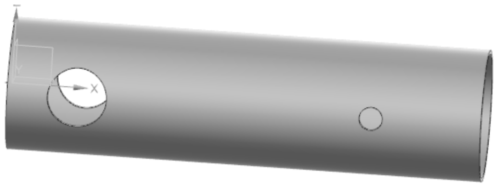
xxii. Cyclic Control Push- Pull Tubes

Material: 6061- T6 Aluminum Tubes (refer xvi for threading)

xxiii. Torque Tube

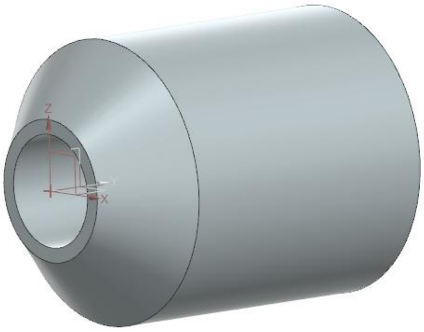
Material: 6061- T6 Aluminum

xxiv. Inner Reinforcement Tube



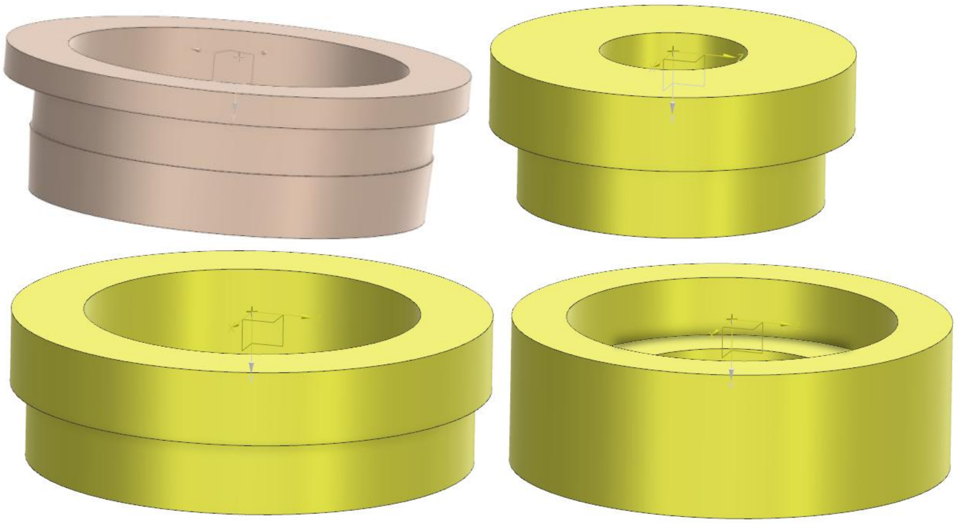
Material: 6061- T6 Aluminum

xxv. Spacer



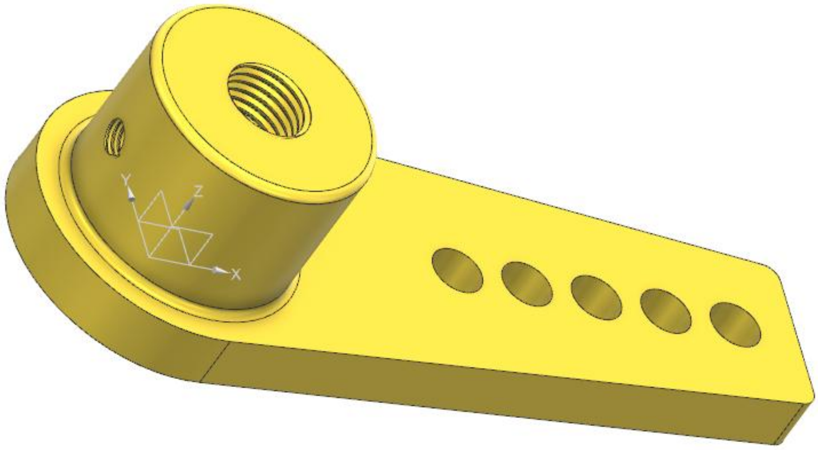
Material: Aluminum

xxvi. Nylon Bushings



Material: Nylon

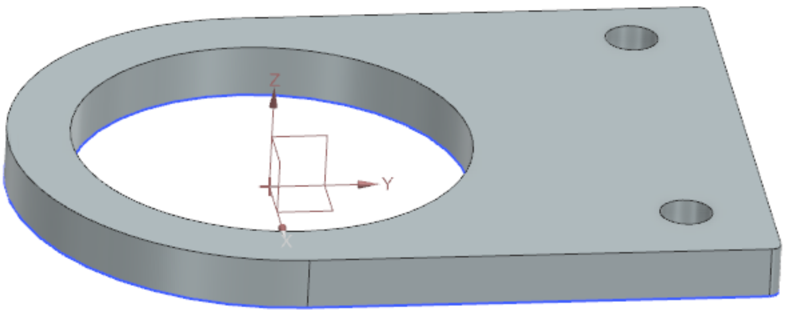
xxvii. Lever



Material:2024- T3/ 6063- T6 Aluminum
Threading-

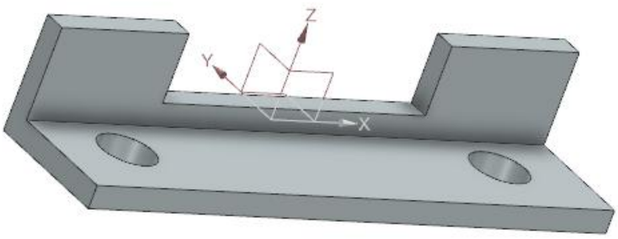
a) Pitch= 0.05in/ 1.27mm
   Major Diameter= 0.4375in/ 11.1125mm
   Minor Diameter=0.3834in/ 9.7384mm
   Tap Drill- 0.390in/ 9.91mm
b) Pitch= 0.03125in/ 0.79375mm
   Major Diameter= 0.19in/ 4.826mm
   Minor Diameter= 0.1562in/ 3.9667mm
   Tap Drill= 0.159in/ 4.10mm

xxviii. Torque Tube Bracket



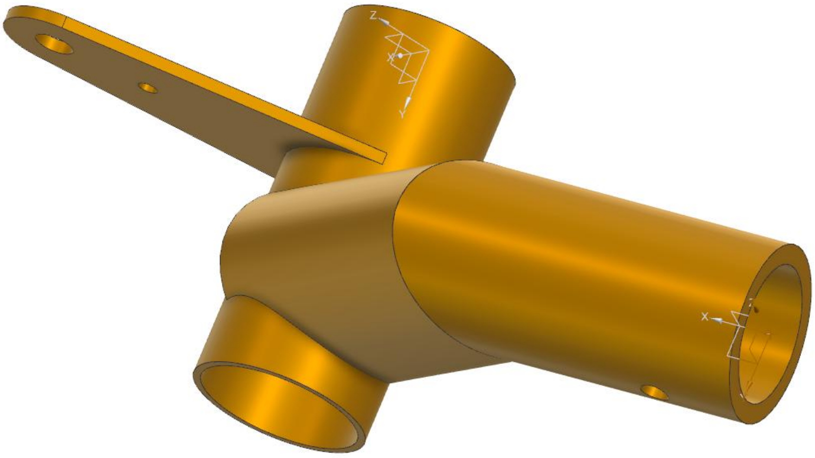
Material: 2024- T3 Aluminum

xxix. Lateral Stop



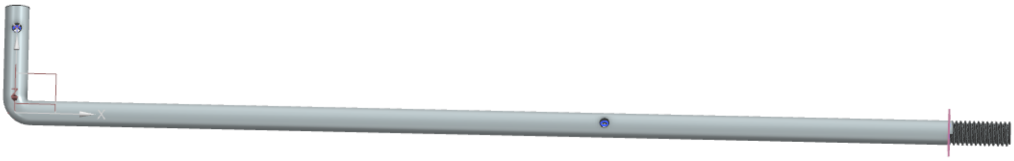
Material: 4130 Steel

xxx. Cyclic Control Stick Holder



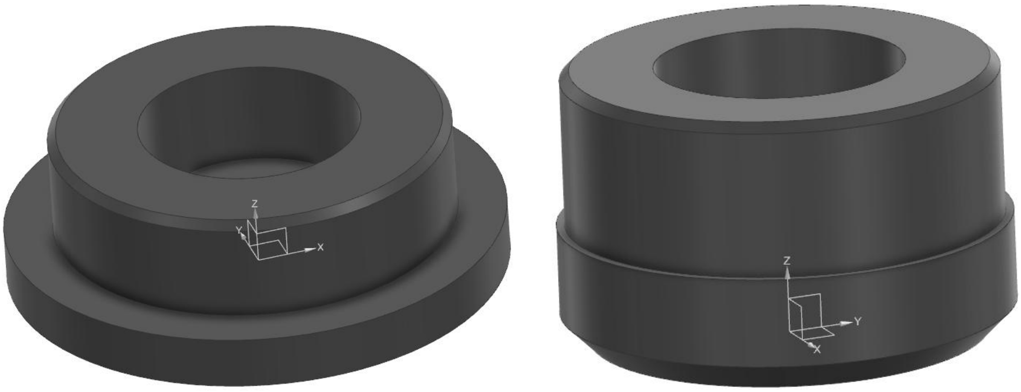
Material: 4130 Steel
Joints are welded.

xxxi. Limiting Bar



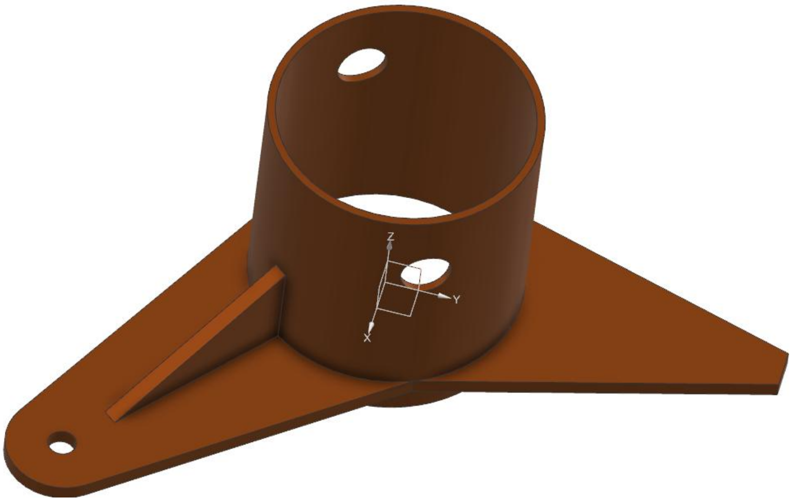
Material: Steel
Threading-
Pitch= 0.0227in/ 0.5773mm
Major Diameter= 0.125in/ 3.175mm
Minor Diameter= 0.1004in/ 2.5501mm
Shank Diameter= 0.125in/ 3.175mm

xxxii. Stick Holder Caps



Material: Aluminum

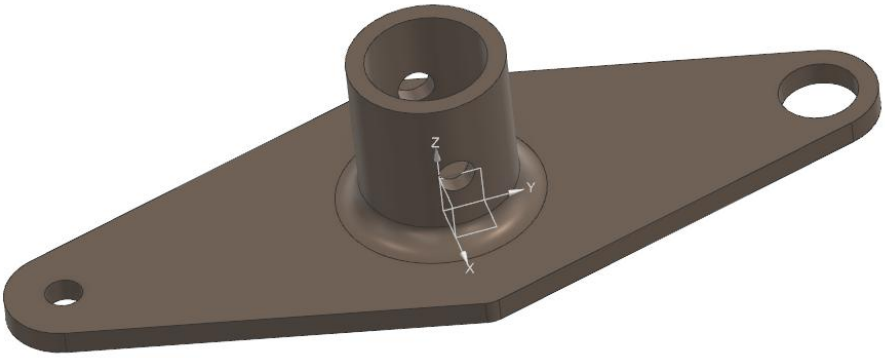
xxxiii. Travel Limiting Stop



Material: 4130 Steel

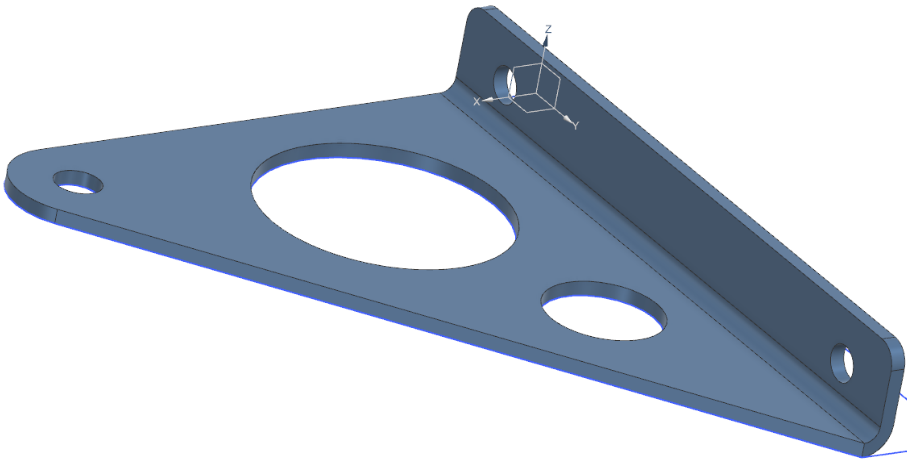
Joints are welded.

xxxiv. Throttle Lever



Material: 4130 Steel. Joints are welded.

xxxv. Upper Cyclic Bellcranks Mounting Bracket



Material: 4130 Steel

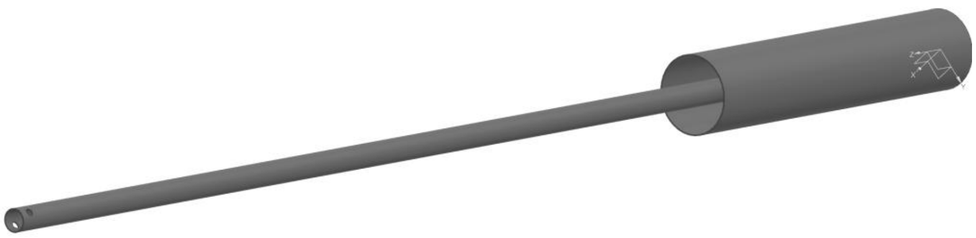
xxxvi. Cyclic Control Stick

Material: 4130 Steel

xxxvii. Collective Control Stick

Material: 4130 Steel Tubing. Joints are welded.
Threading-
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill- 0.213in/ 5.4mm

xxxviii. Hand Throttle Control



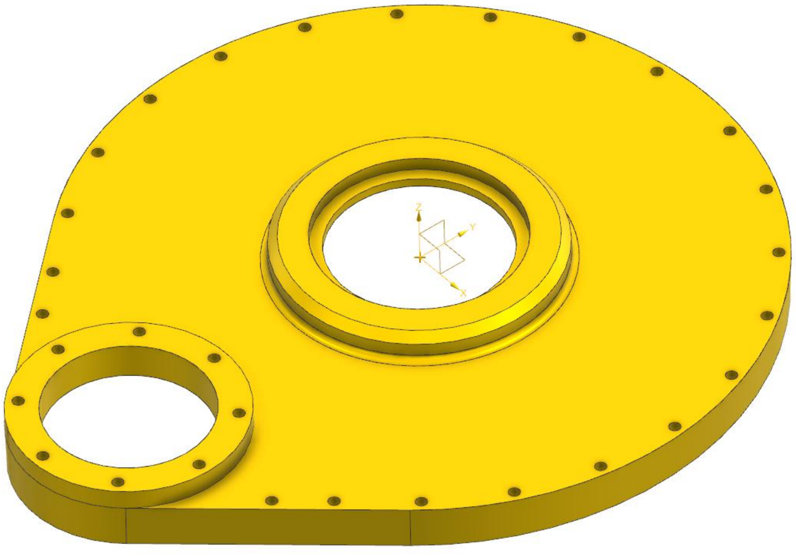
Material: 4130 Steel Tube. The end part is welded.

xxxix. Pin

Material: 4140 Steel

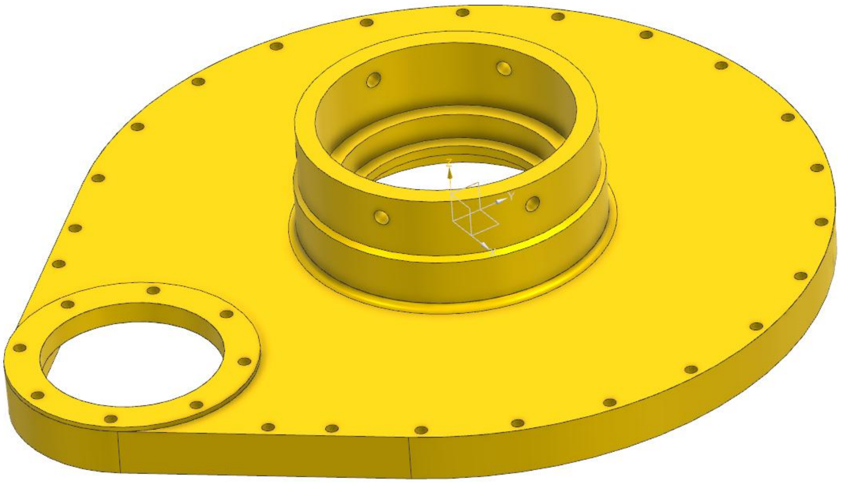
8. Main Gearbox

i. Gearbox Housing (Upper Half)



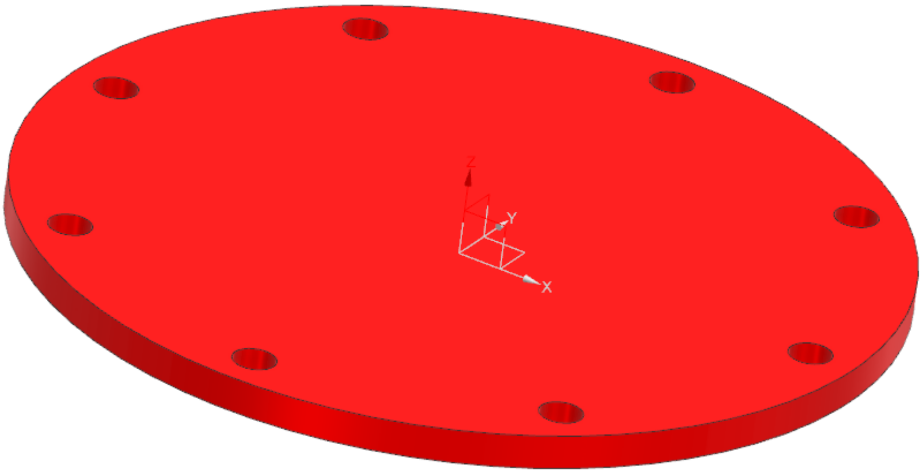
Material: Aluminum Casting

ii. Gearbox Housing (Lower Half)



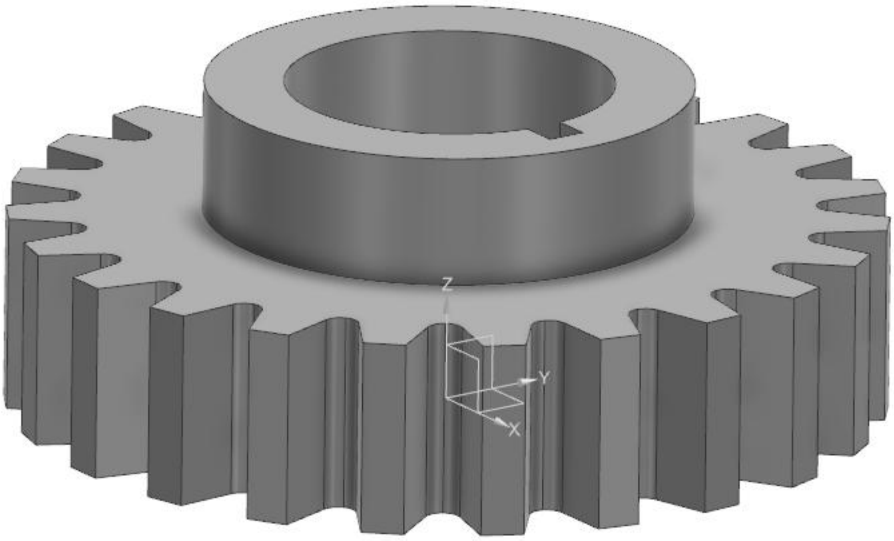
Material: Aluminum Casting

iii. Gearbox Lid

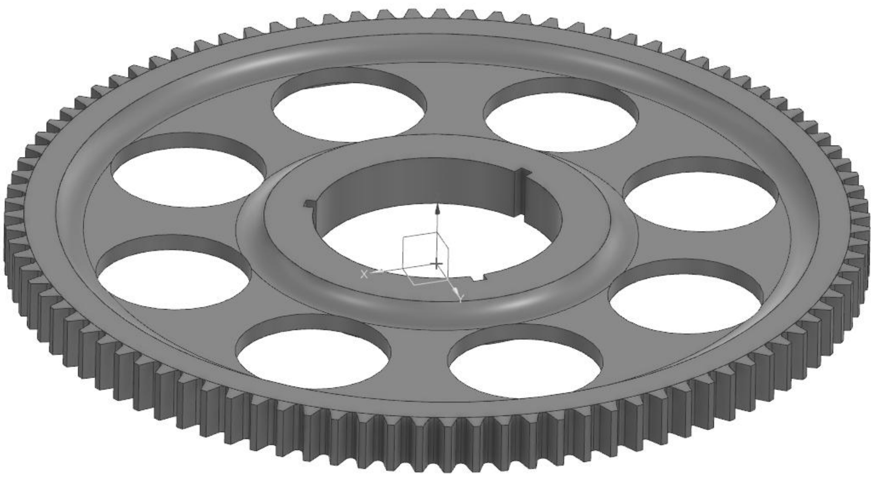


Material: Aluminum

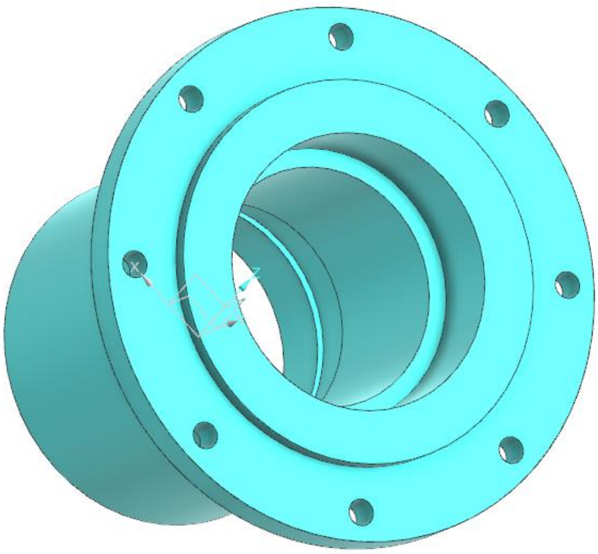
iv. Pinion



v. Main Gear

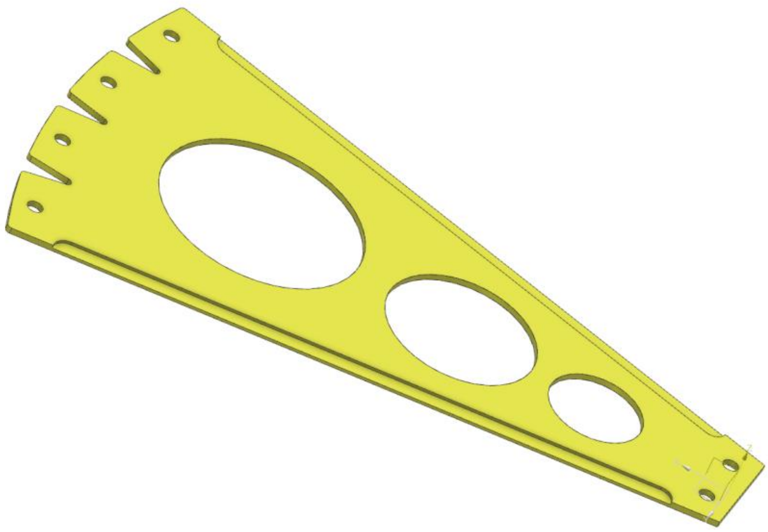


vi. Pinion Housing



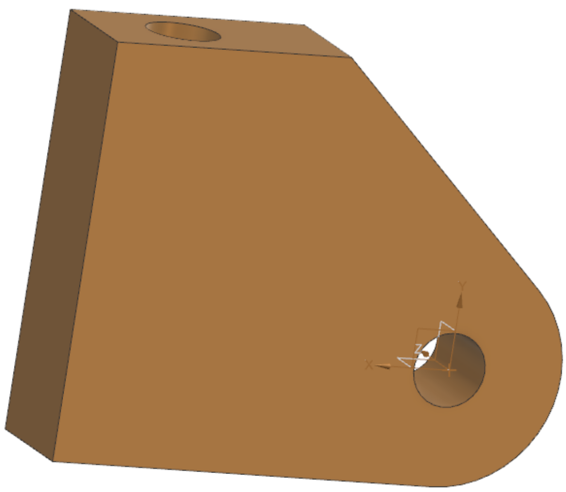
Material: Aluminum

vii. Front Bracket



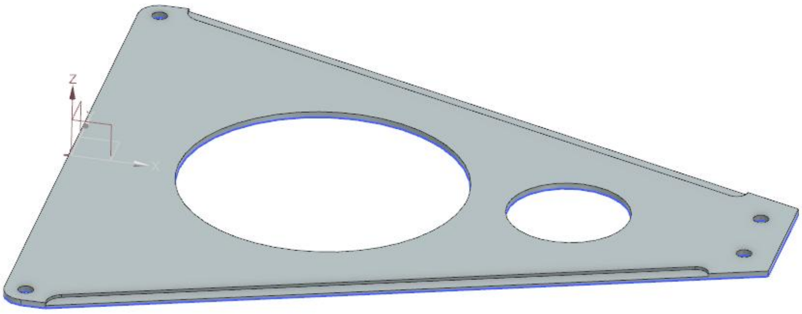
Material: 4130 Steel

viii. Bellcrank Bracket



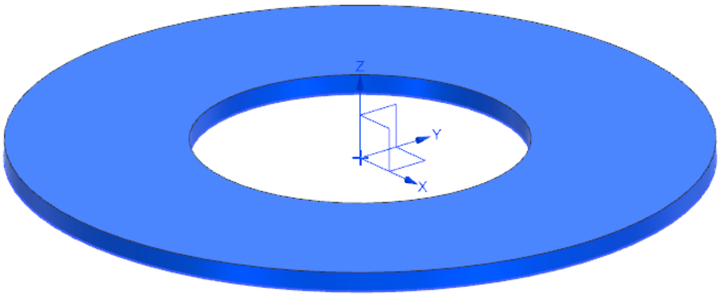
Material: 2024- T3 Aluminum

ix. Rear Bracket



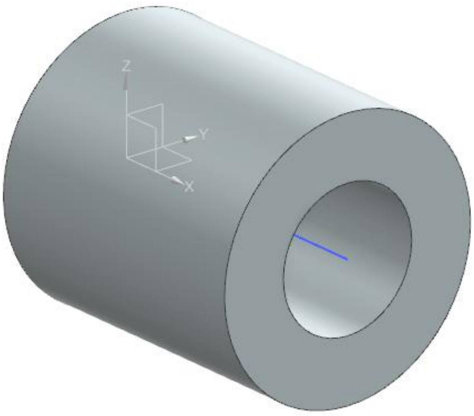
Material: 4130 Steel

x. Gasket



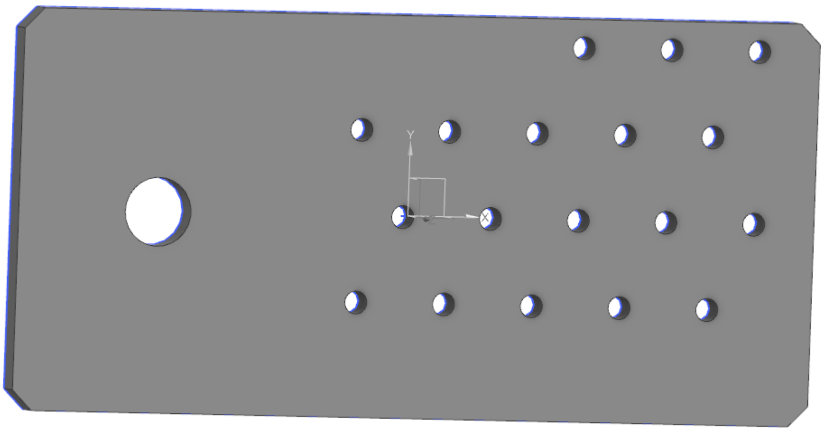
9. Main Rotor Blade

i. Steel Bushing



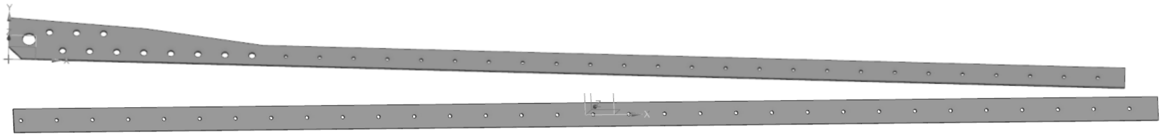
Material: Steel

ii. Blade Root Reinforcement Plate



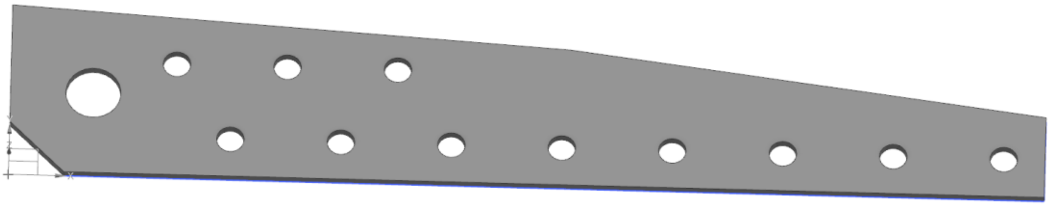
Material: 2024- T3 Aluminum

iii. Inner Reinforcement



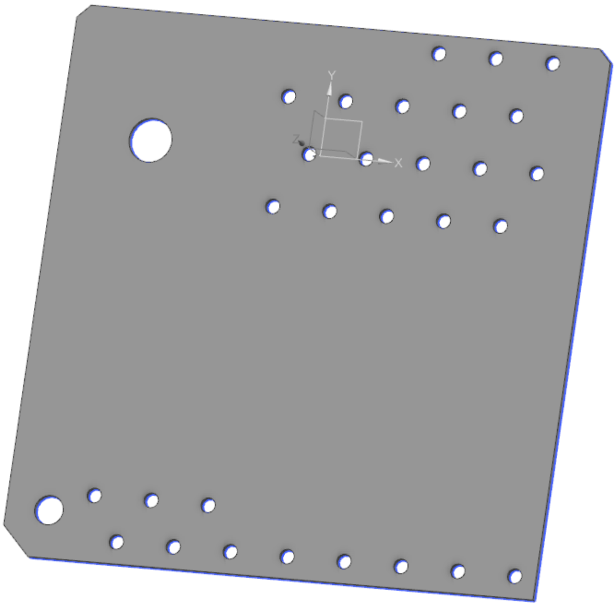
Material: 4130 Steel

iv. Upper Reinforcement



Material: 2024- T3 Aluminum

v. Lower Blade Plate



Material: Steel

10. Tail Rotor

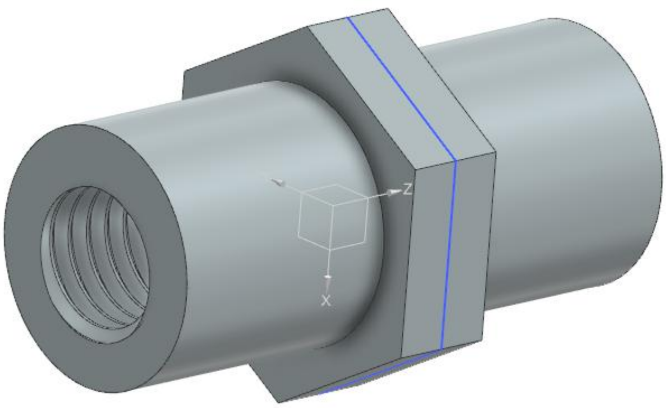
i. Pitch Control Head

Material: 2024- T3 Aluminum

ii. Steel Pin

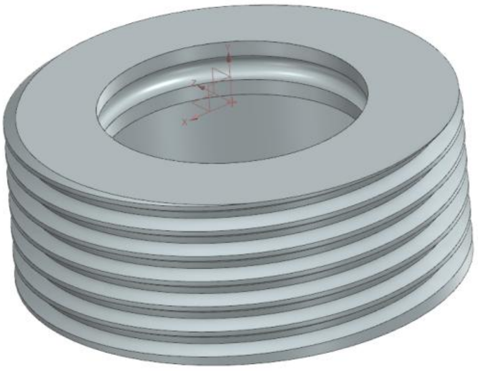
Material: Steel

iii. Pitch Control Linkage



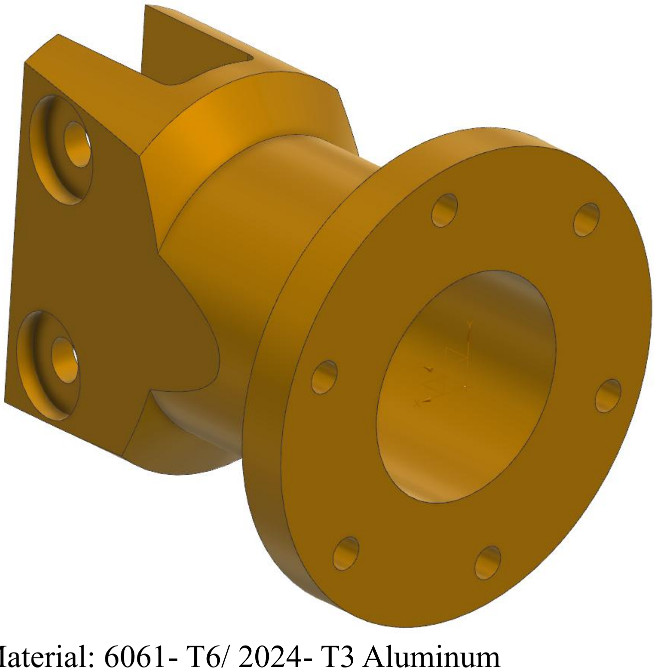
Material: Aluminum Bar
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill- 0.213in/ 5.4mm

iv. Blade Grip Nut



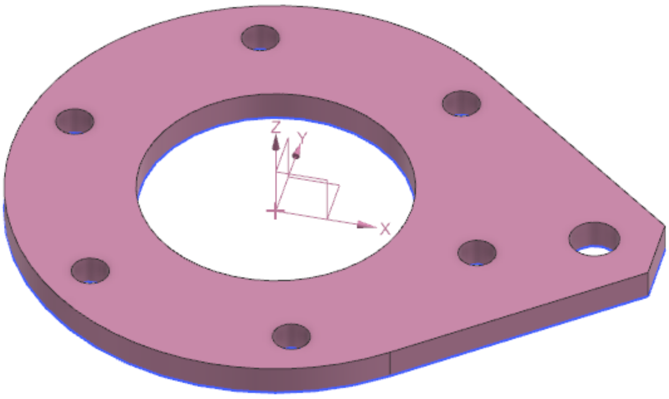
Material: 4130 Steel
Threading-
Pitch= 0.0833in/ 2.09mm
Major Diameter= 1.3750in/ 34.925mm
Minor Diameter= 1.2848in/ 32.6346mm
Shank Diameter= 1.3667in/ 34.7133mm

v. Tail Blade Grip



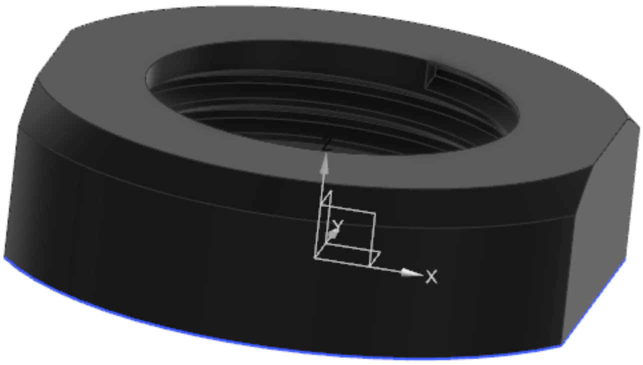
Material: 6061- T6/ 2024- T3 Aluminum
Threading-
Pitch= 0.0833in/ 2.09mm
Major Diameter= 1.3750in/ 34.925mm
Minor Diameter= 1.2848in/ 32.6346mm
Tap Drill= 1.2969in/ 32.94mm

vi. Pitch Control Horn



Material: 2024- T3 Aluminum

vii. Axle Nut



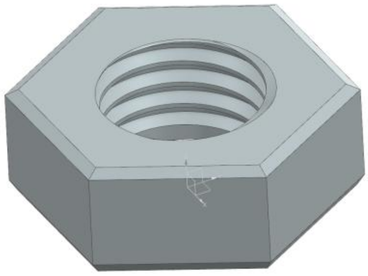
Material: Steel
Threading-
Pitch= 0.0556in/ 1.4111mm
Major Diameter= 0.625in/ 15.875mm
Minor Diameter= 0.5649in/ 14.3474mm
Tap Drill= 0.5781in/ 14.50mm

viii. Retainer Nut

Material: Steel
Threading-
Pitch= 0.5556in/ 1.4111mm

Major Diameter= 0.5625in/ 14.2875mm
Minor Diameter= 0.5024in/ 12.7599mm
Tap Drill= 0.5156in/ 12.90mm

ix. Axle Pivot Screws and Nuts



Material: Steel
Pitch= 0.0417in/ 1.0583mm
Major Diameter= 0.3125in/ 7.9375mm
Minor Diameter=0.2674in/ 6.7918mm
Tap Drill- 0.2720in/ 6.9mm
Shank Diameter= 0.3125in/ 7.9375mm

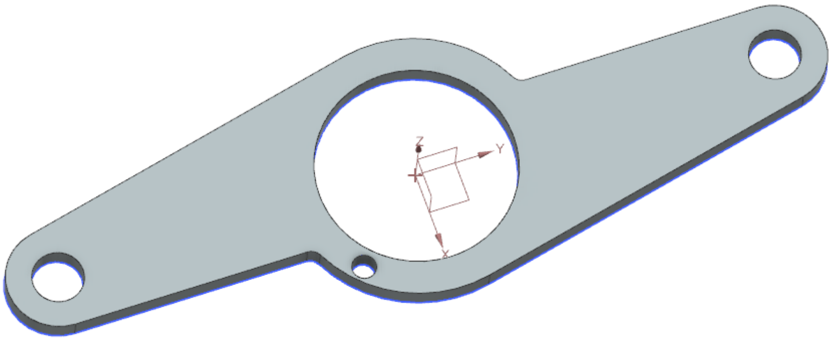
x. Tail Rotor Hub

Material: 2024- T3 Aluminum

xi. Pitch Control Axle

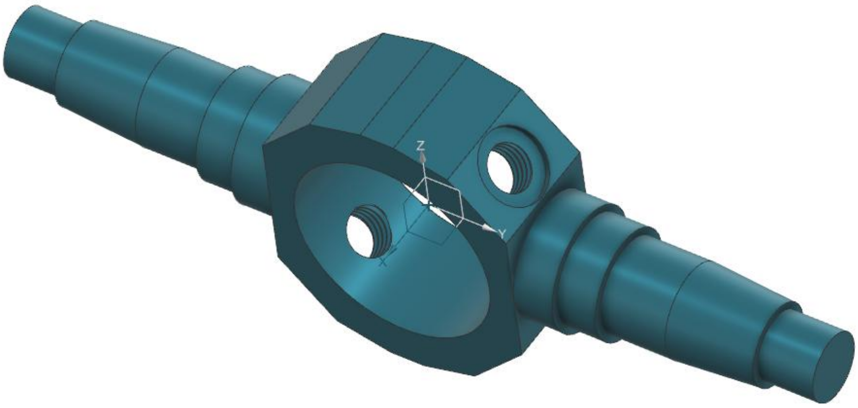
Material: Steel
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill- 0.213in/ 5.4mm

xii. Control Head Guide



Material: Steel

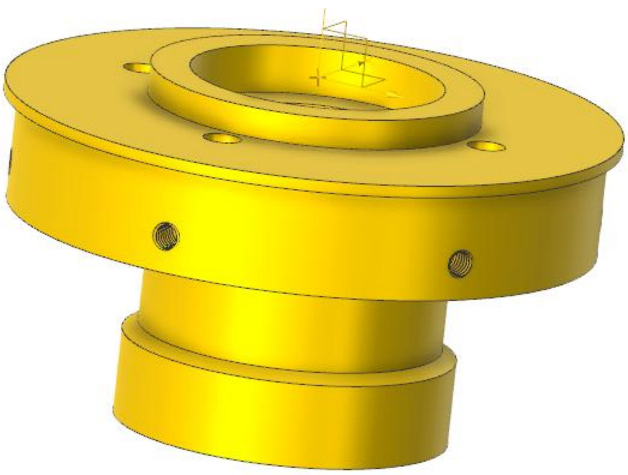
xiii. Axles



Material: 4130 Steel (refer ix for threading)

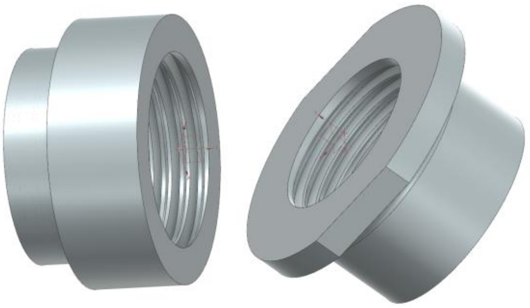
Material: Steel

xiv. Gearbox Rear Cap



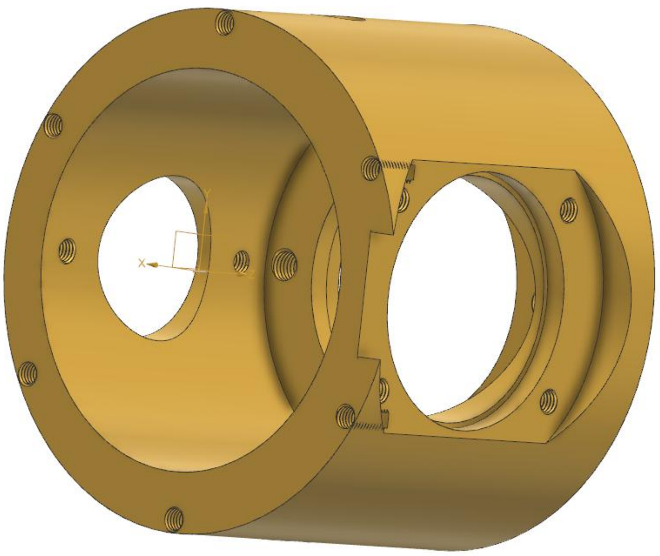
Material: Aluminum
Threading-
Pitch= 0.03125in/ 0.79375mm
Major Diameter= 0.19in/ 4.826mm
Minor Diameter= 0.1562in/ 3.9667mm
Tap Drill= 0.1590in/ 4.04mm

xv. Nuts



Material: Steel
Threading-
a) Pitch= 0.0556in/ 1.4111mm
   Major Diameter= 0.625in/ 15.875mm
   Minor Diameter= 0.5649in/ 14.3474mm
   Tap Drill= 0.5781in/ 14.50mm
b) Pitch= 0.0625in/ 1.5875mm
   Major Diameter= 0.75in/ 19.05mm
   Minor Diameter= 0.6823in/ 17.3315mm
   Tap Drill= 0.689in/ 17.50mm

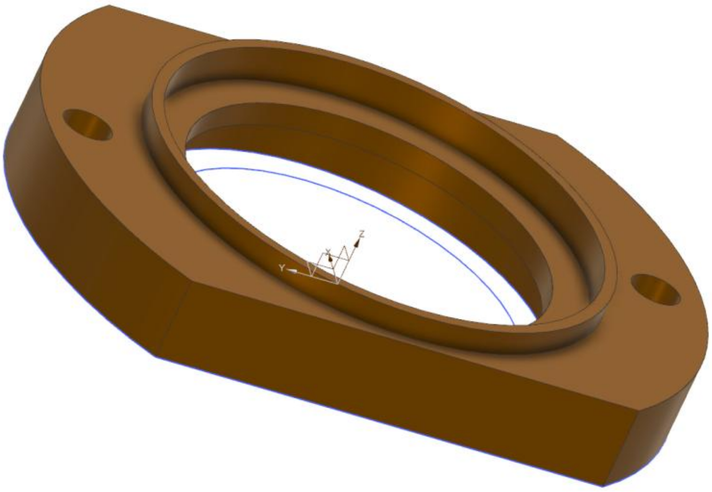
xvi. Gearbox



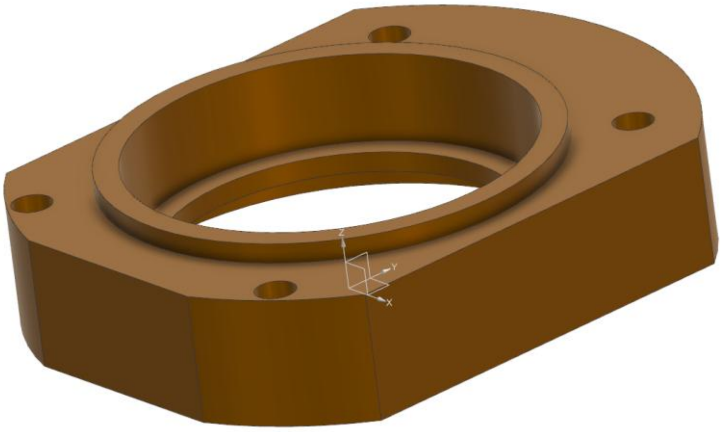
Material: Aluminum
Threading-
a) Pitch= 0.03125in/ 0.79375mm
   Major Diameter= 0.19in/ 4.826mm
   Minor Diameter= 0.1562in/ 3.9667mm
   Tap Drill= 0.1590in/ 4.04mm
b) Pitch= 0.0357in/ 0.907mm

Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill- 0.213in/ 5.4mm

xvii. Bearing Covers



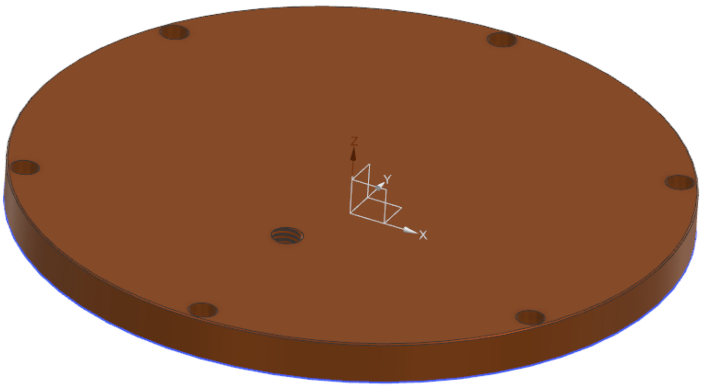
EE5



16005

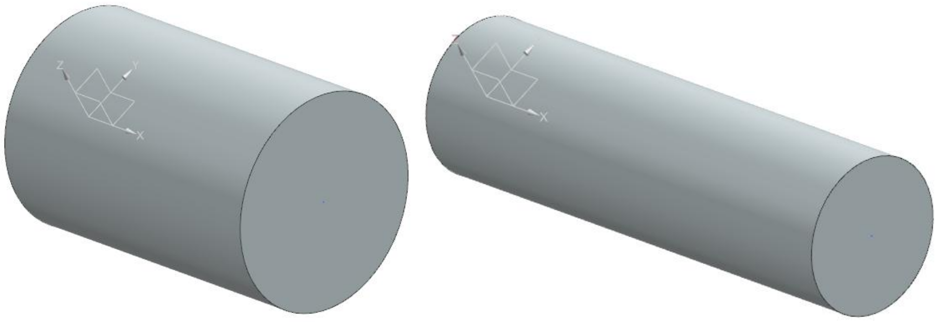
Material: Aluminum

xviii. Tail Rotor Lid



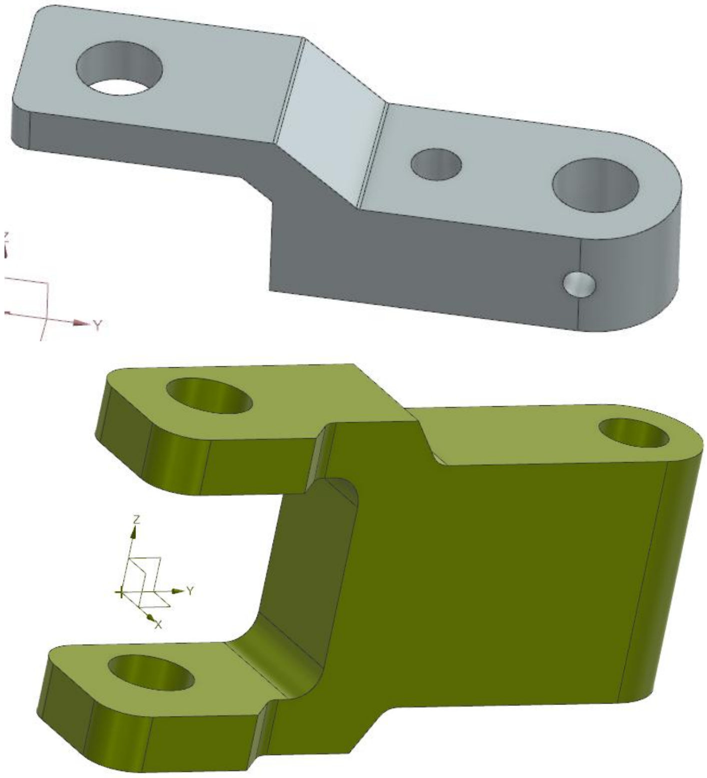
Material: Aluminum (refer xvi for threading)

xix. Gears Steel Bushings



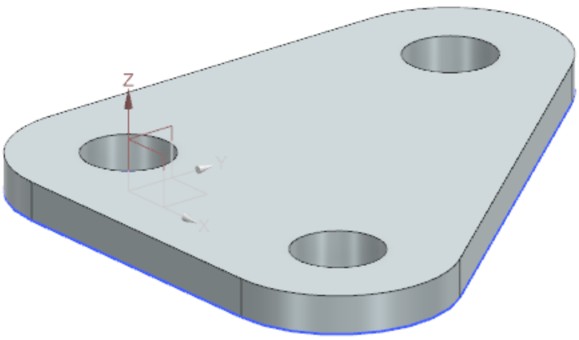
Material: Steel

xx. Scissors



Material: 2024- T3 Aluminum

xxi. Pulley Arm

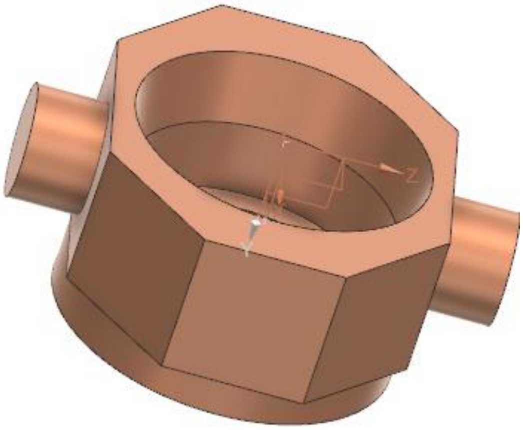


Material: Steel

xxii. Lever

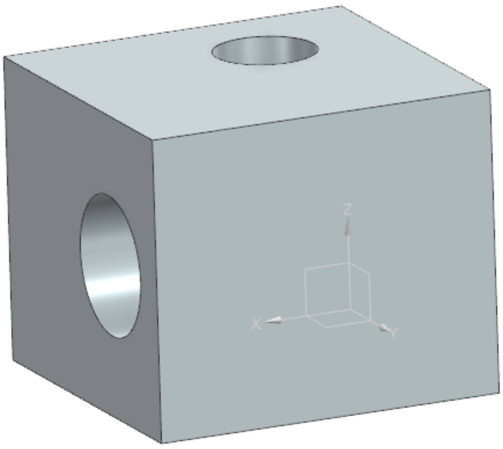
Material: Steel

xxiii. Bearing Housing



Material: Aluminum

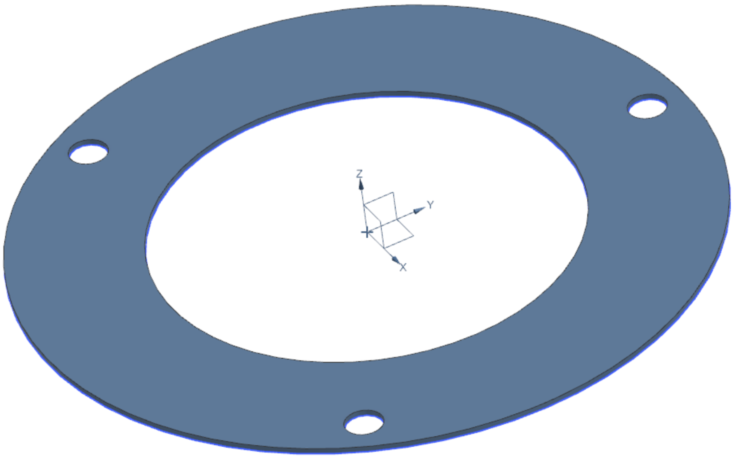
xxiv. Scissors Hinge



Material: 2024- T3 Aluminum

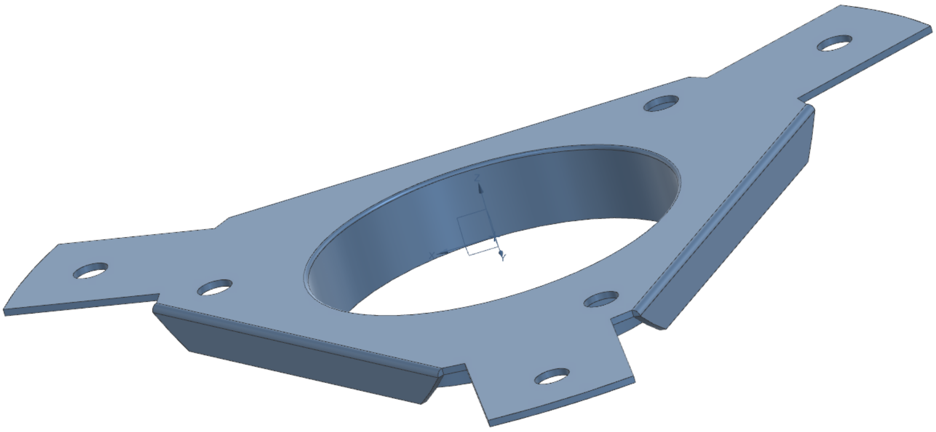
11.Tail Boom

i. Retaining Washer



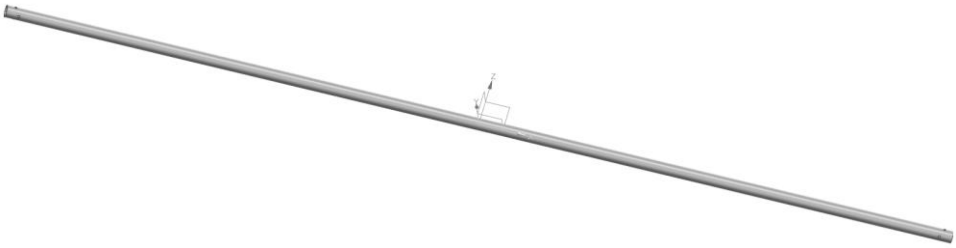
Material: Aluminum

ii. Internal Bracket



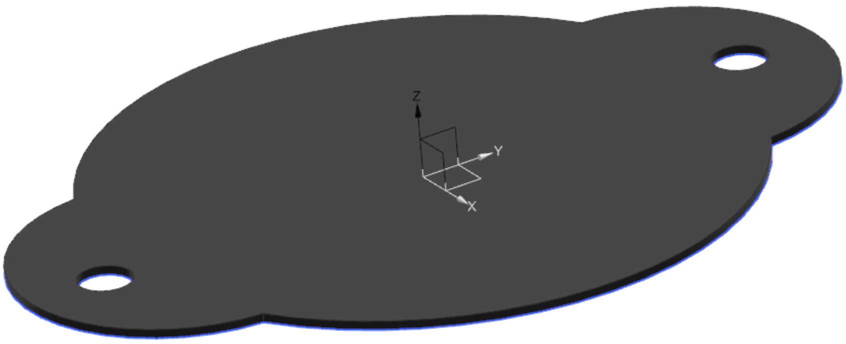
Material: 2024- T3 Aluminum

iii. Shaft



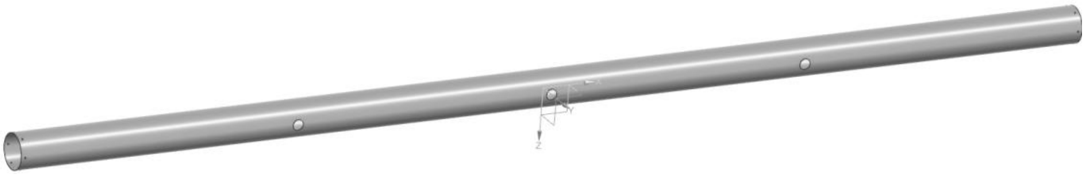
Material: 6061- T6 Aluminum

iv. Inspection Hole Cover



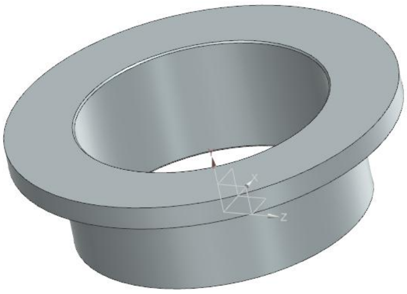
Material: Aluminum

v. Tail Boom Tube



Material: 6061- T6 Aluminum

vi. Bushing

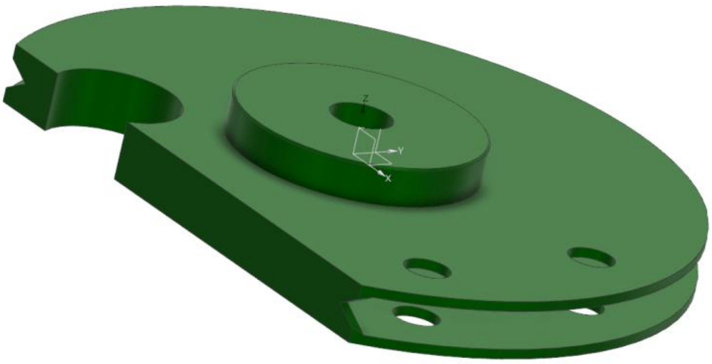


Material: Black Grilon (with Molykote)

vii. Raceway

Material: 8620 Steel (Case Hardened)

viii. Control Pulley



Material: 2024- T3 Aluminum

12. Pedals

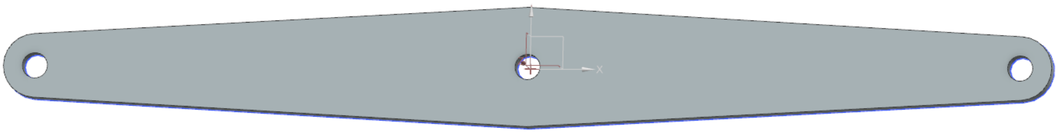
i. Pedals Axle

Material: 4130 Steel. Joints are welded.
Threading-
Pitch= 0.0357in/ 0.907mm
Major Diameter= 0.2500in/ 6.350mm
Minor Diameter=0.2113in/ 5.367mm
Tap Drill- 0.213in/ 5.4mm

ii. Directional Control Pedals

Material: 4130 Steel

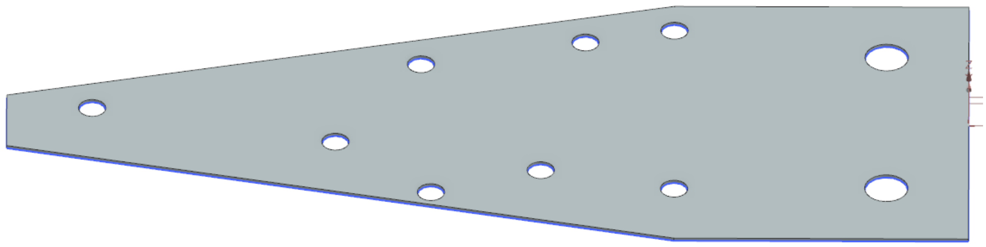
iii. Main Lever



Material: 2024- T3 Aluminum

13. Tail Rotor Blade

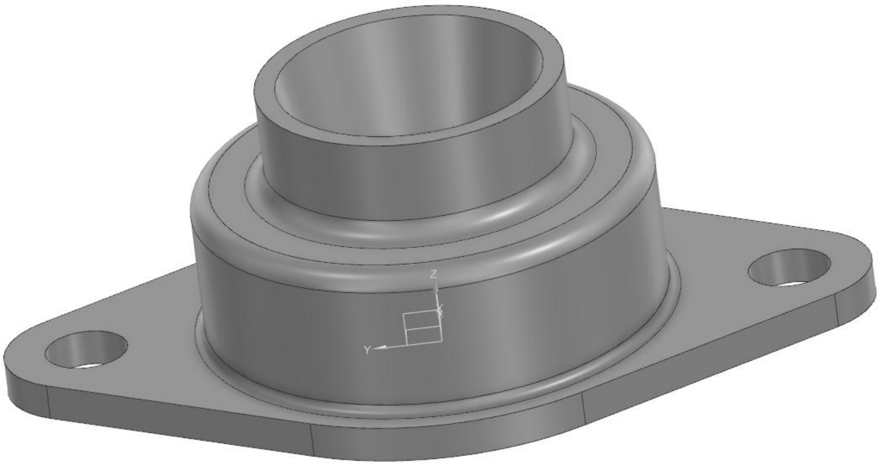
i. Reinforcement



Material: 2024- T3 Aluminum

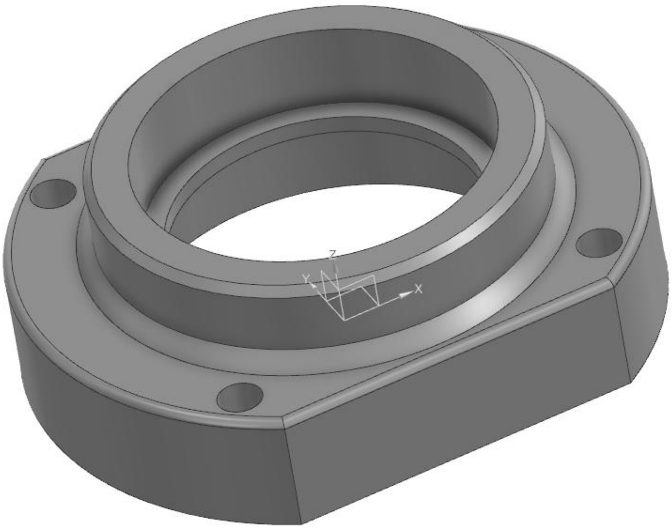
14. Central Gearbox

i. Long Axle Coupling



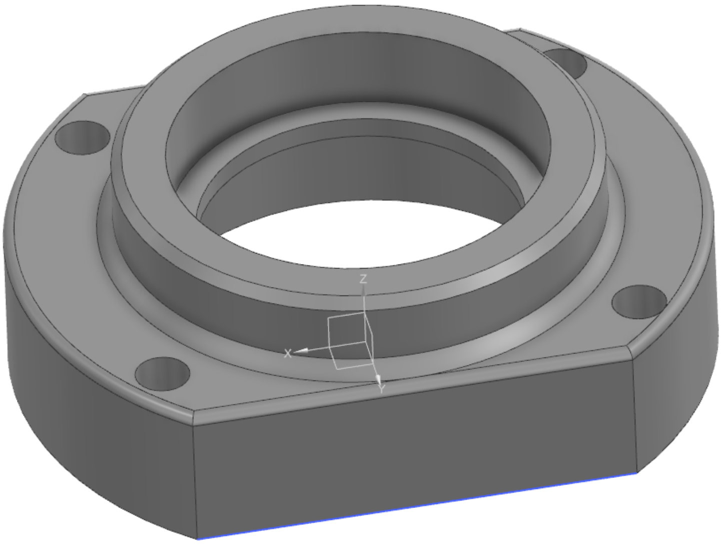
Material: Steel

ii. Upper Bearing Cover



Material: Aluminum

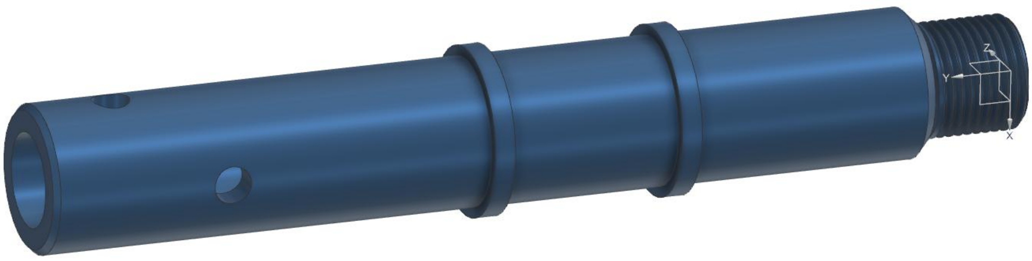
iii. Lower Bearing Cover



Material: Aluminum

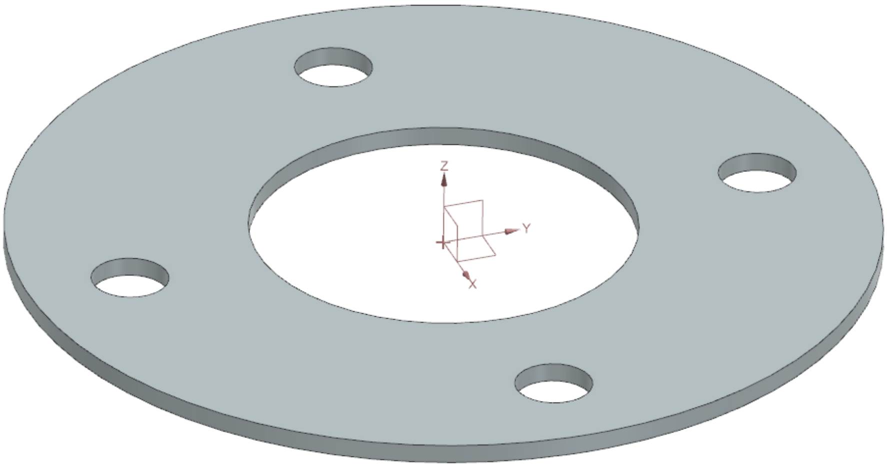
15. Transmission Components

i. Pinion Axle

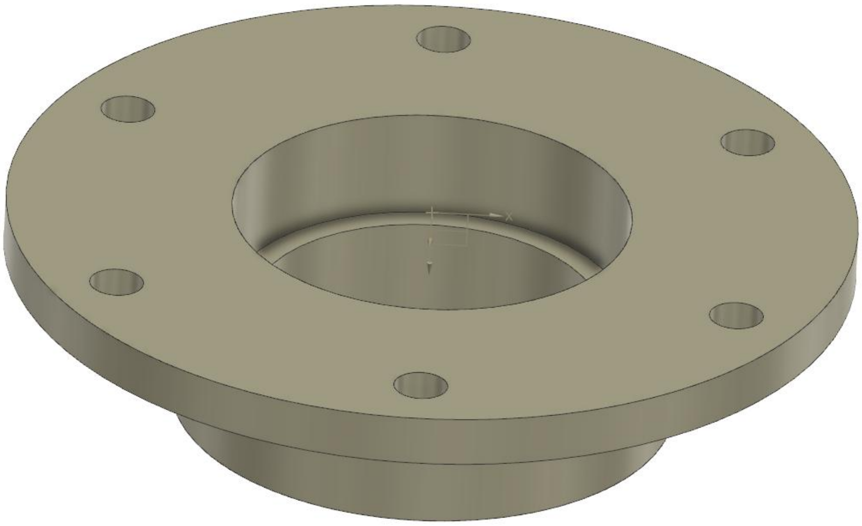


Threading-
Pitch= 0.0625in/ 1.5875mm
Major Diameter= 0.75in/ 19.05mm
Minor Diameter= 0.6823in/ 17.3315mm
Shank Diameter= 0.75in/ 19.05mm

ii. Connecting Ring

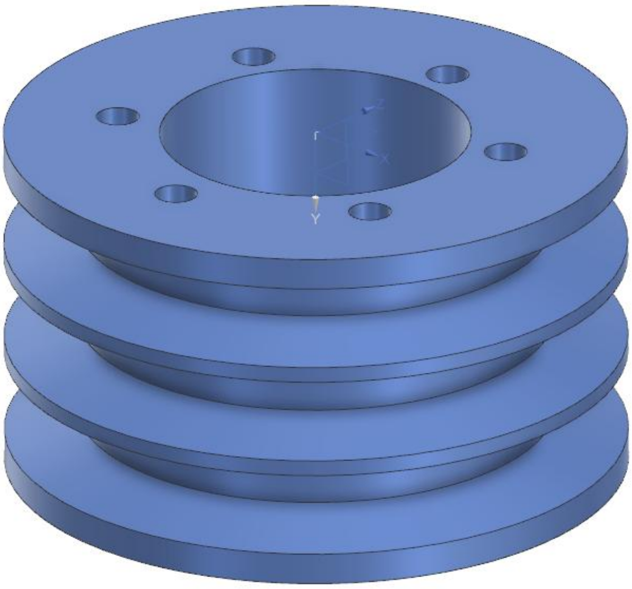


iii. Pulley Support



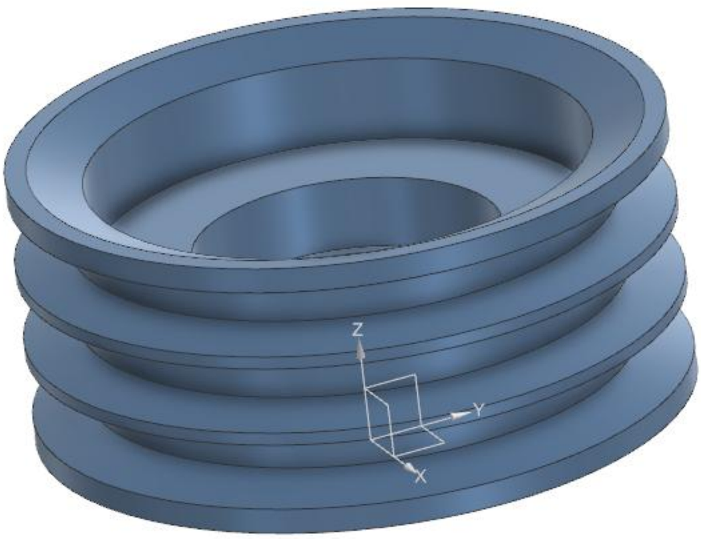
Material: Aluminum

iv. Engine Pulley



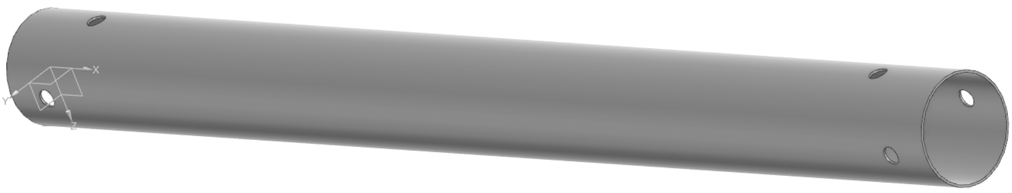
Material: Aluminum

v. Freewheel



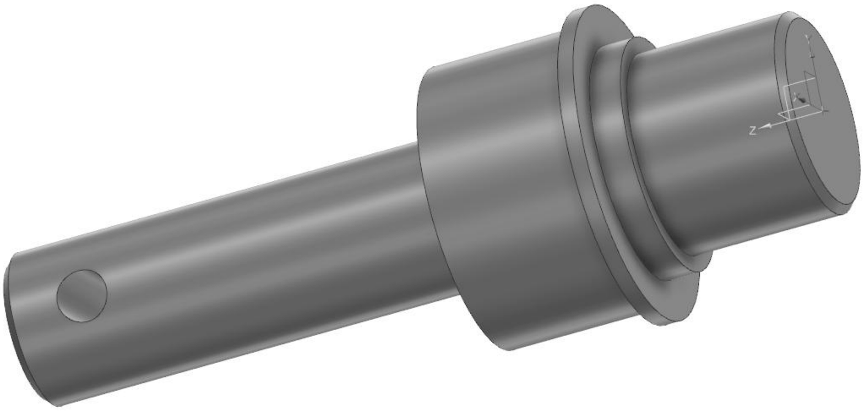
Material: Steel

vi. Transmission Shaft



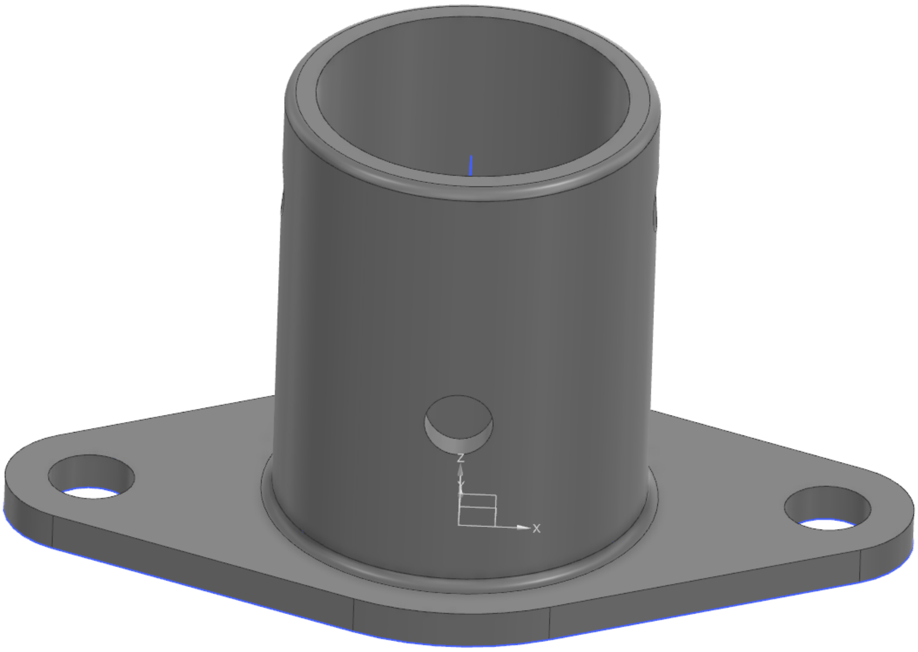
Material: 4130 Steel Tube

vii. Freewheel Axle



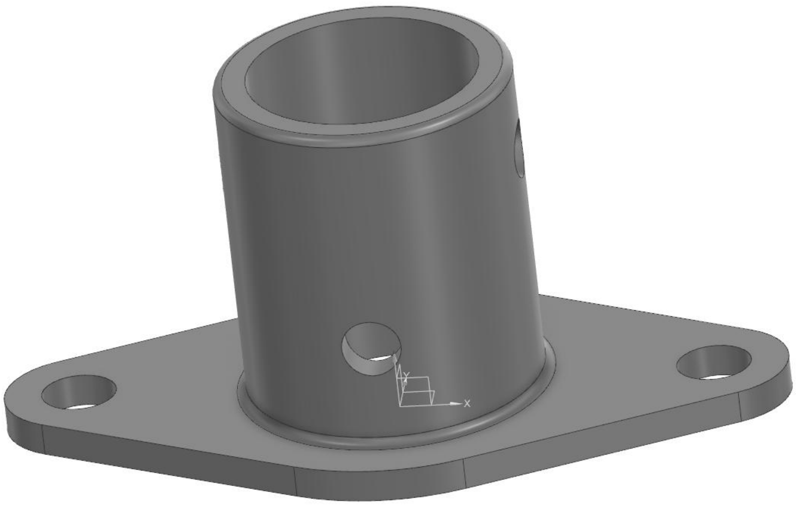
Material: Steel

viii. Shaft Coupling



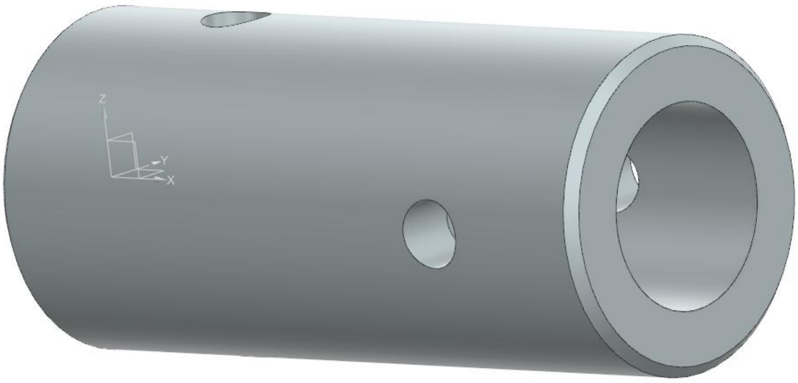
Material: Steel

ix. Pinion Axle Coupling



Material: Steel

x. Freewheel Axle Coupling



Material: Steel

The End